



MMAUG / AI & DEVOPS BOOTCAMP

Introduction to AI Concepts and Tools

A practical foundation for understanding how AI works, where it fits, which tools matter and how to use it responsibly

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1 October 2026

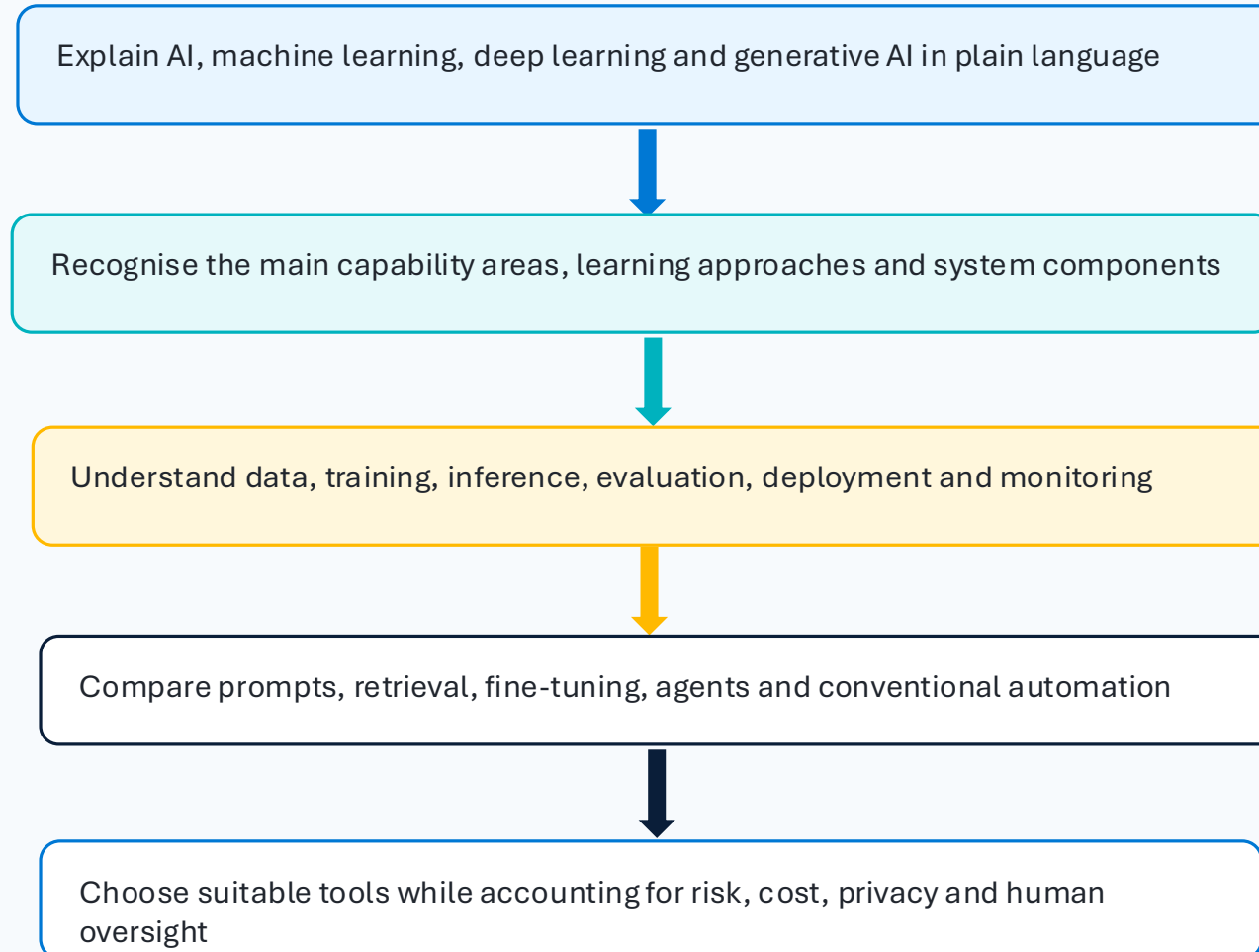
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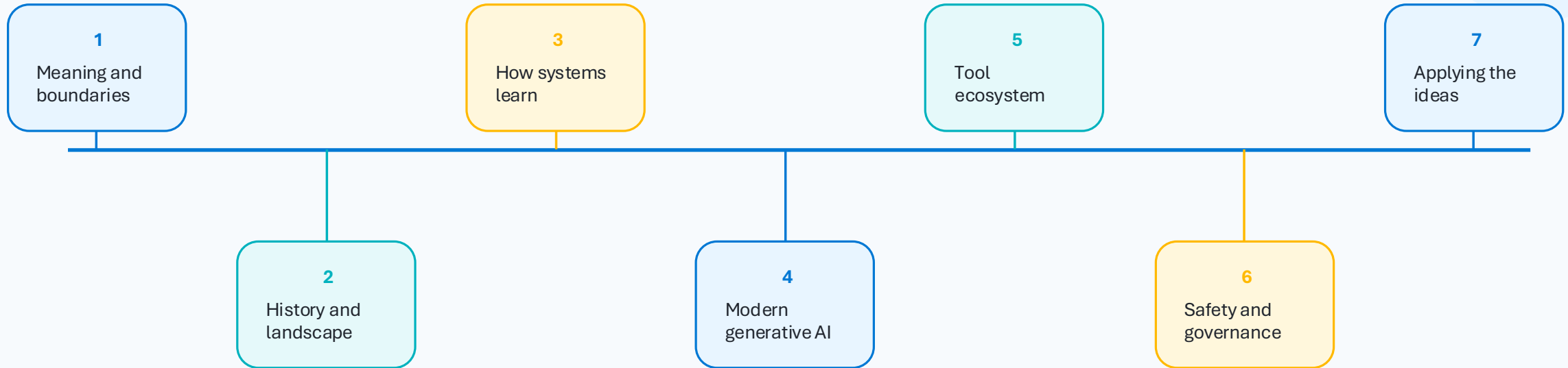
02 / LEARNING OUTCOMES

This session builds a usable map of AI



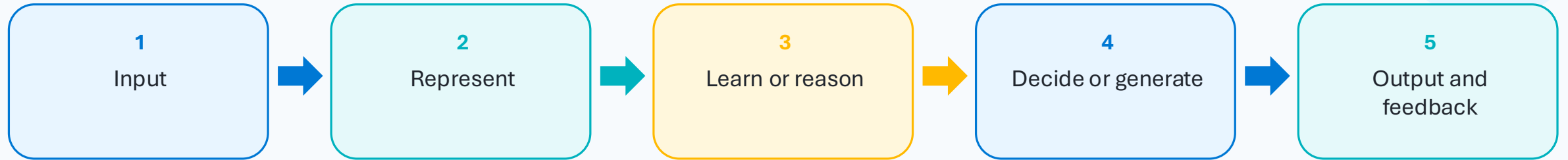
03 / SESSION MAP

Seven connected views of the field



04 / WHAT IS AI?

AI turns information into useful machine behaviour

**Key message**

An AI system performs a task that normally requires perception, prediction, language, reasoning or action.

05 / BOUNDARIES

AI imitates selected capabilities, not an entire human mind

A AI can support

- Pattern recognition
- Prediction and classification
- Language and media generation
- Optimisation and recommendation
- Planning within defined constraints



B AI does not automatically provide

- Human understanding or consciousness
- Common sense in every situation
- Truth, fairness or accountability
- Guaranteed explanations
- Independent responsibility

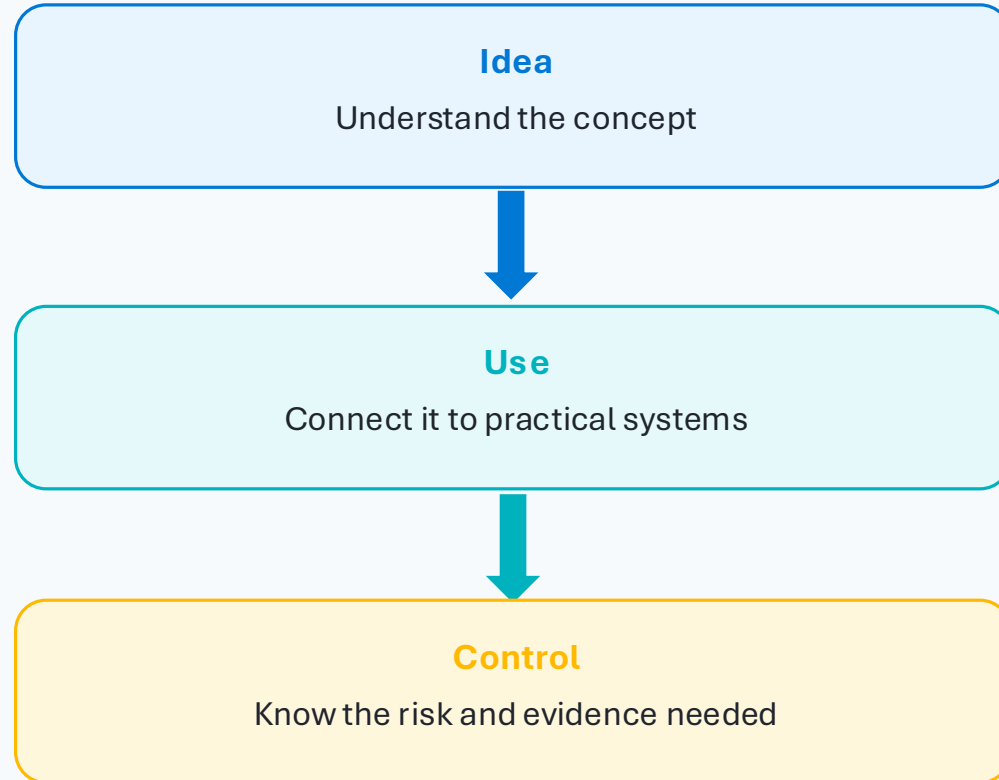
Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

SECTION / 01

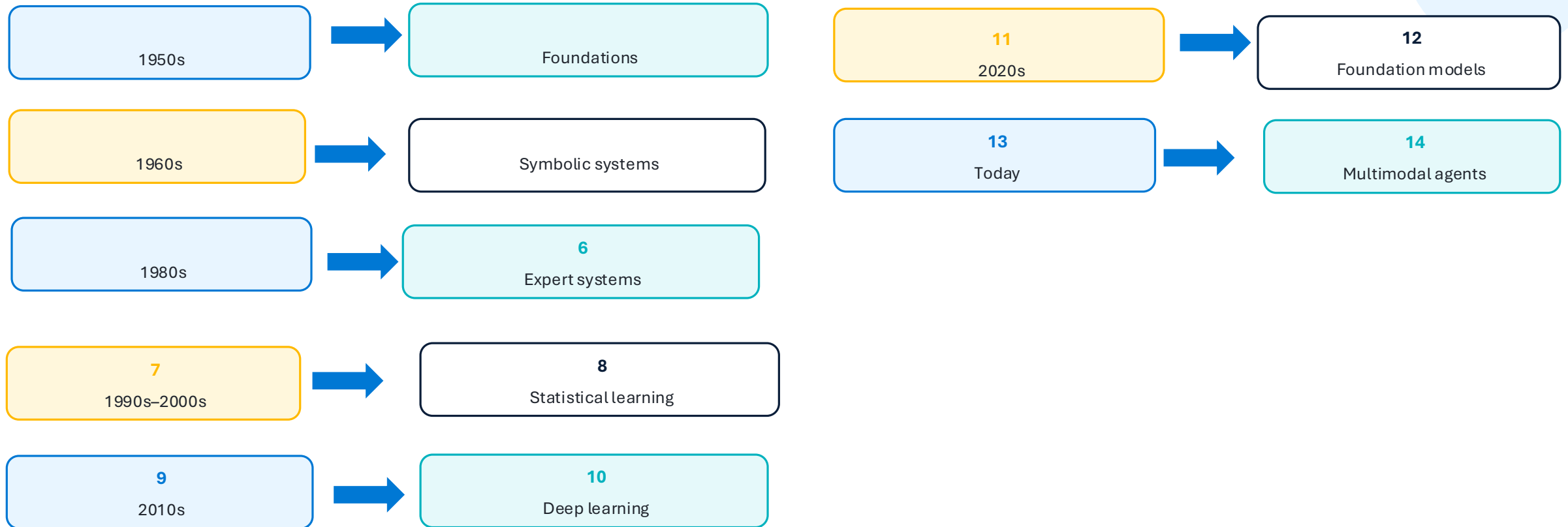
How AI evolved

Ideas, computing power, data and changing expectations shaped the field



07 / AI HISTORY

The field developed through distinct eras



08 / CYCLES OF PROGRESS

AI progress arrived in waves

A Growth periods

- New algorithms attracted attention
- More compute enabled larger experiments
- Data availability improved results
- Successful applications increased investment



B AI winters

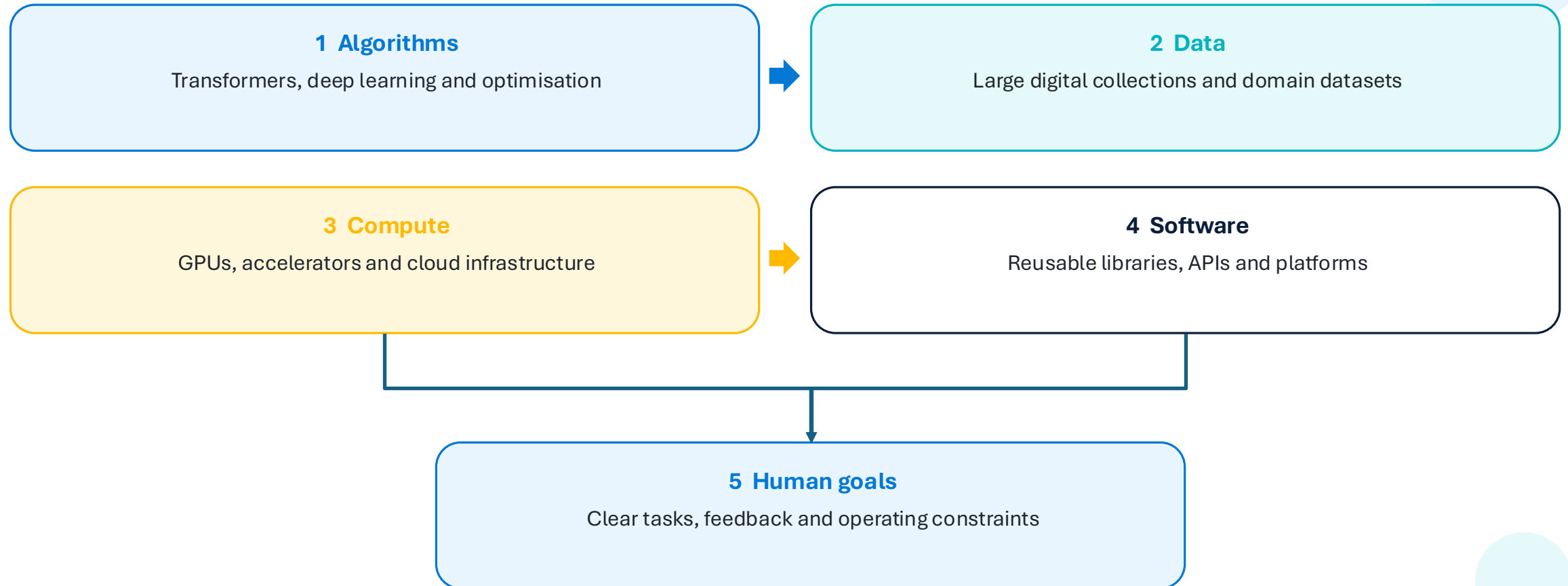
- Expectations exceeded capability
- Funding and public interest declined
- Hardware or data limits became visible
- Research continued beneath the slowdown

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

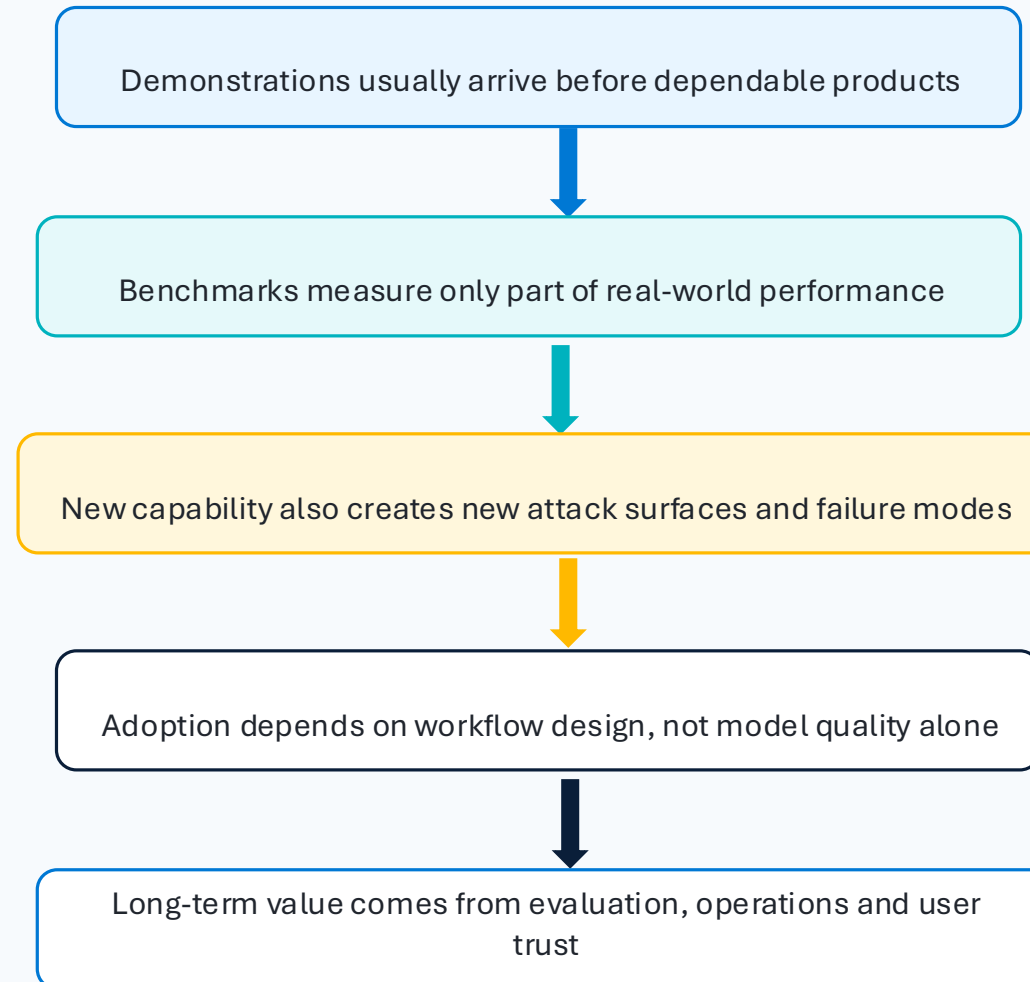
09 / WHY NOW?

Modern AI combines five enabling forces



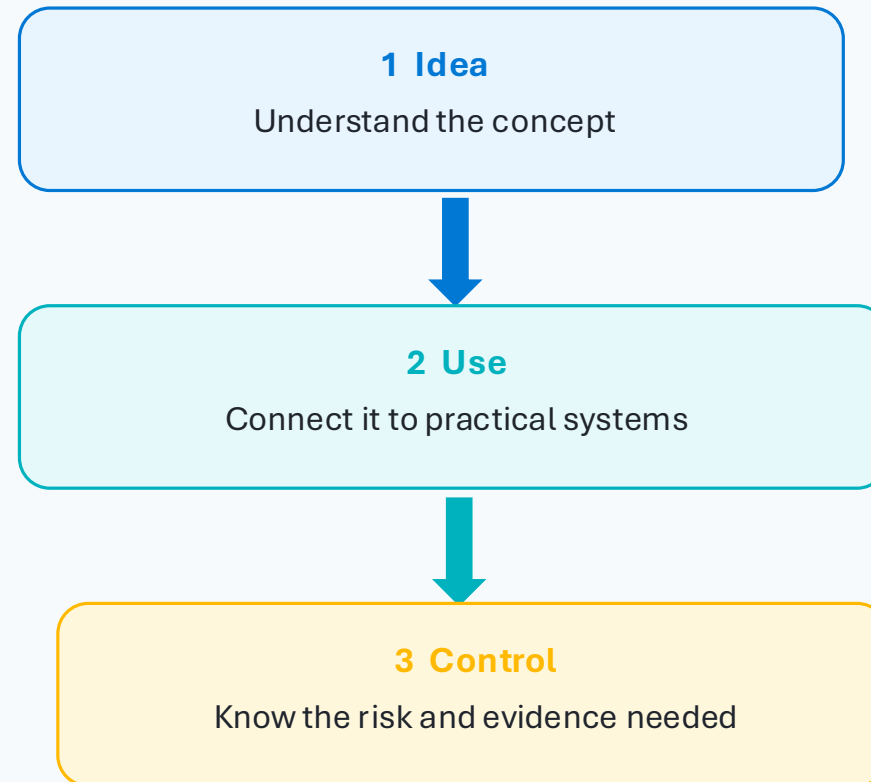
10 / HISTORY LESSON

Capability grows faster than reliability



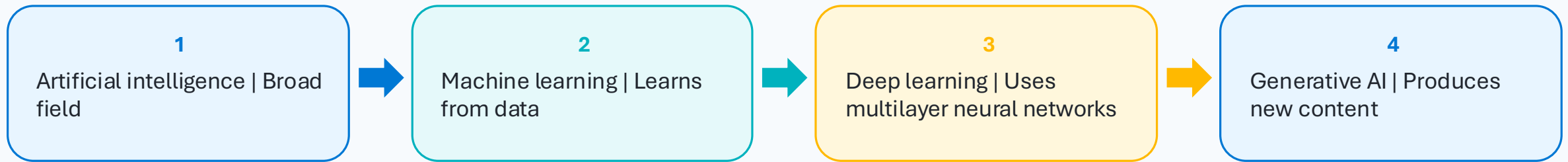
The landscape of AI

AI is an umbrella containing different methods, capabilities and application areas



12 / CORE RELATIONSHIP

AI, machine learning and deep learning overlap

**Key message**

The terms describe nested areas, but real products often combine several methods.

13 / TWO TRADITIONS

Symbolic and learning-based AI solve problems differently

A Symbolic AI

- Uses explicit rules, logic and knowledge
- Often easier to trace
- Works well when rules remain stable
- Struggles with ambiguity and perception



B Learning-based AI

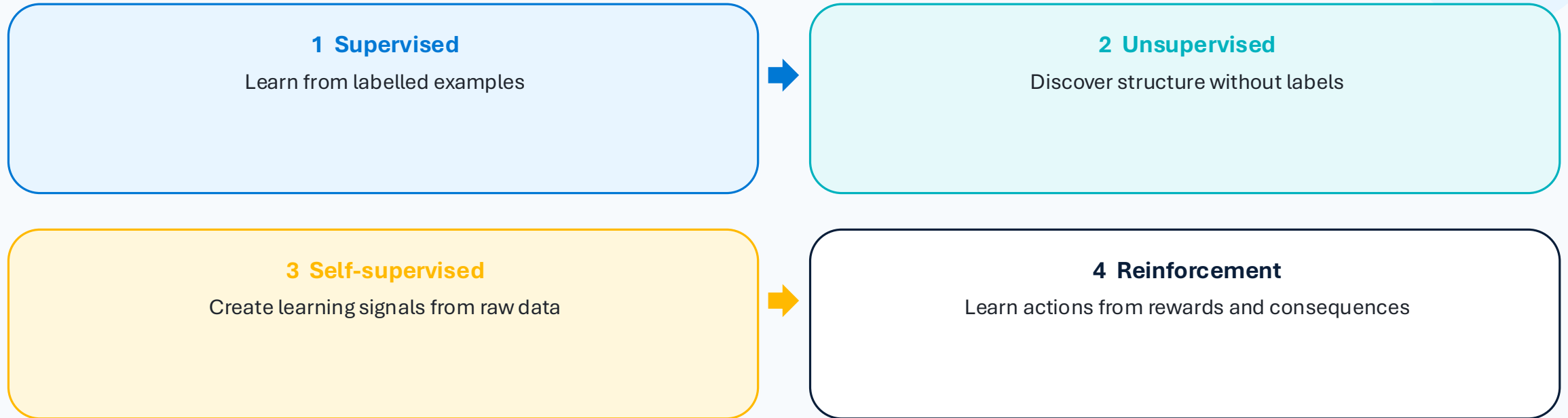
- Learns statistical patterns from examples
- Adapts to complex data
- Can generalise beyond training examples
- May be difficult to explain

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

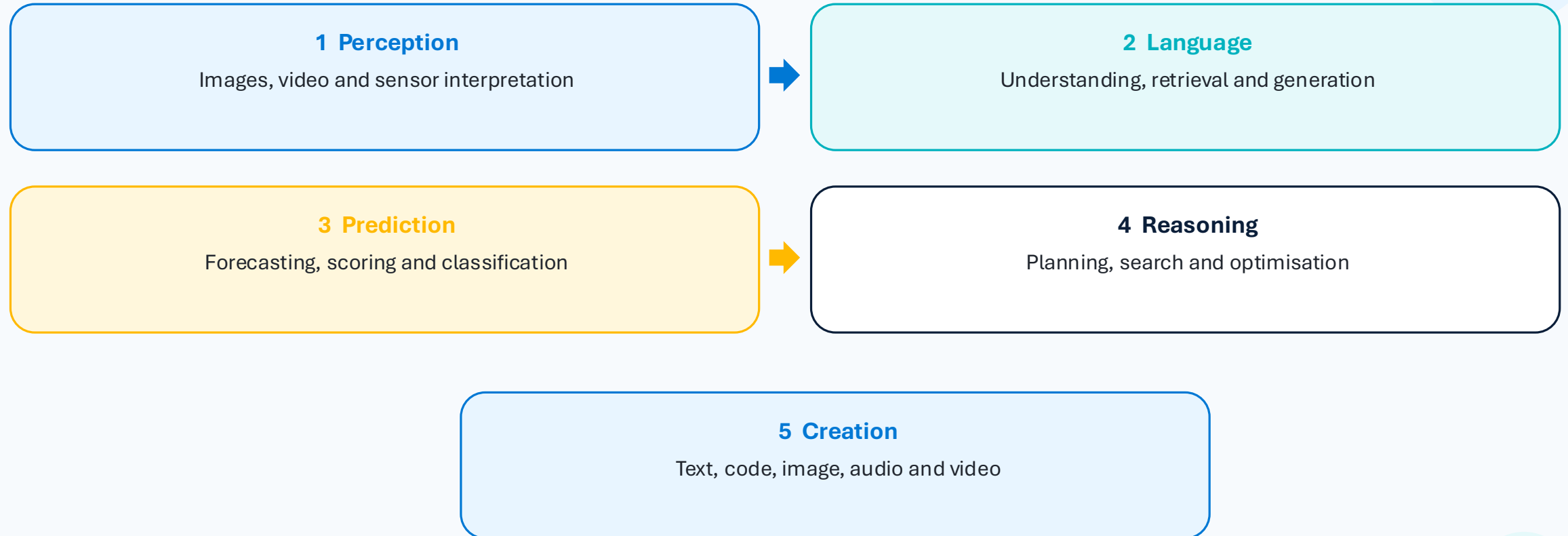
14 / LEARNING PARADIGMS

Different learning approaches answer different questions



15 / CAPABILITY AREAS

AI capabilities span several domains



16 / SCOPE

Narrow AI is today's practical reality

A Narrow AI

- Designed for a defined task or domain
- Can exceed humans on selected benchmarks
- Depends on its training and operating context
- Includes most deployed systems today



B General intelligence

- Would transfer learning broadly across domains
- Would adapt with human-like flexibility
- Remains a research aspiration
- Should not be confused with fluent conversation

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

17 / MODEL PURPOSE

Discriminative and generative models answer different questions

A Discriminative

- Predict a label, score or boundary
- Estimate which category fits the input
- Examples: spam detection and risk scoring
- Usually optimised for a defined target



B Generative

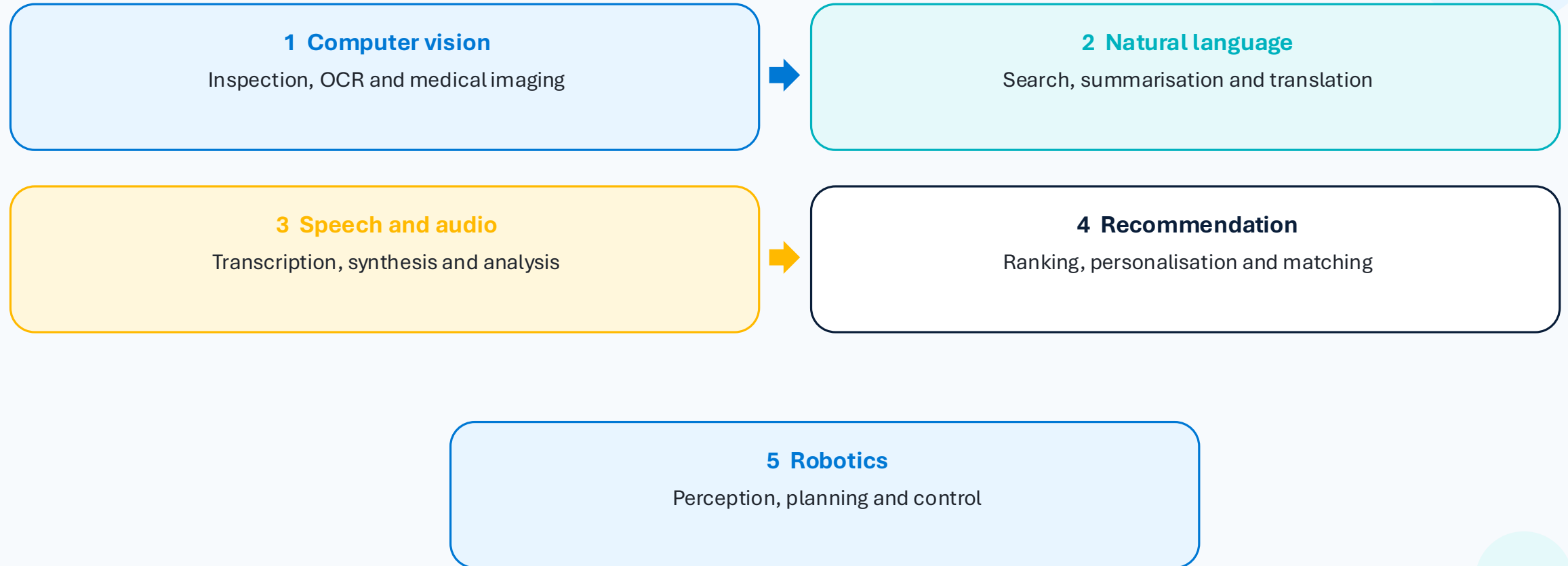
- Model how data is structured
- Produce plausible new samples or sequences
- Examples: text, images, code and audio
- Require controls for truthfulness and safety

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

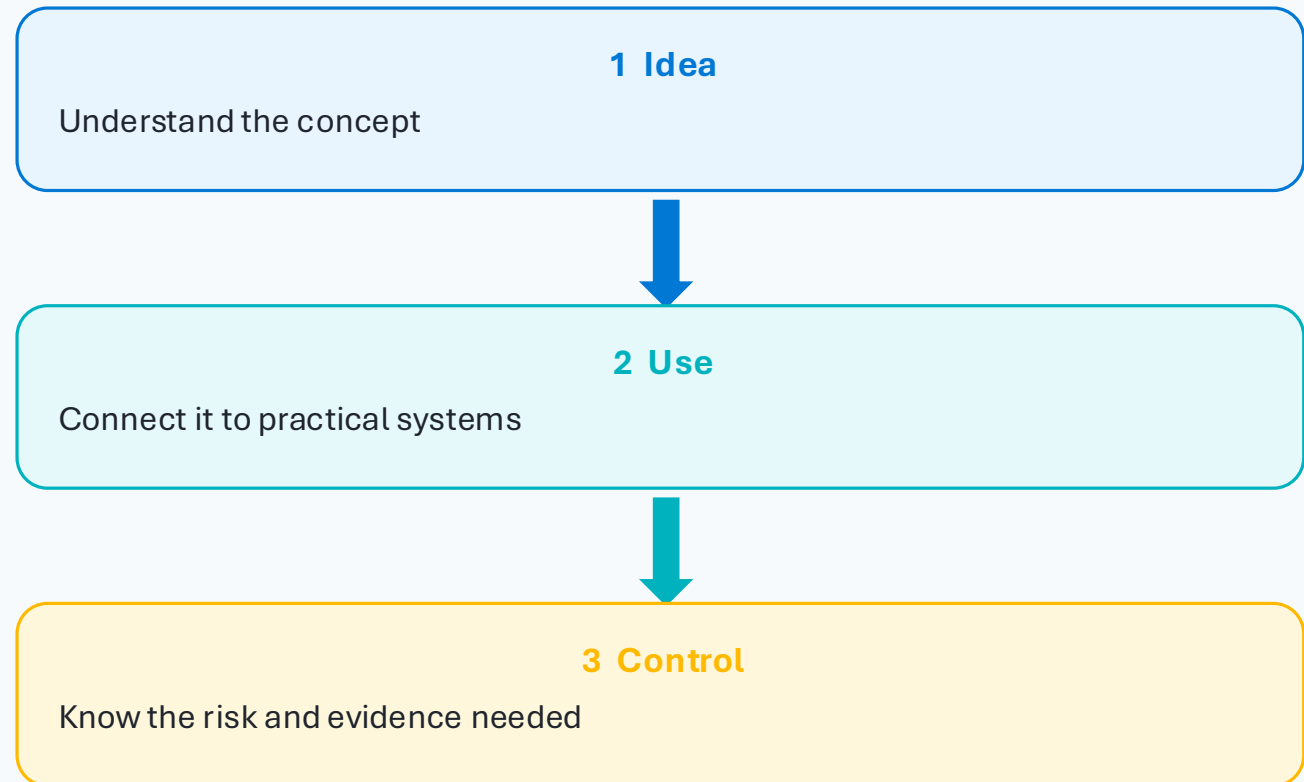
18 / APPLICATION DOMAINS

Common AI applications combine multiple capabilities



Machine learning foundations

Models learn patterns from examples and apply them to new data



Machine learning

D Definition

Machine learning is a way to build software that learns a pattern from examples instead of receiving every rule from a programmer.



M Mechanism

A team supplies data, selects a learning algorithm, trains a model and tests how well it handles unseen examples.



E Example

A bank trains a fraud model on past transactions, then scores each new payment for unusual behaviour.

Note

Typical tasks include classification, regression, clustering, recommendation and anomaly detection.

Deep learning

D Definition

Deep learning is machine learning that uses neural networks with many layers to learn useful representations from complex data.



M Mechanism

Early layers detect simple patterns. Later layers combine those patterns into higher-level features that support a prediction or generation task.



E Example

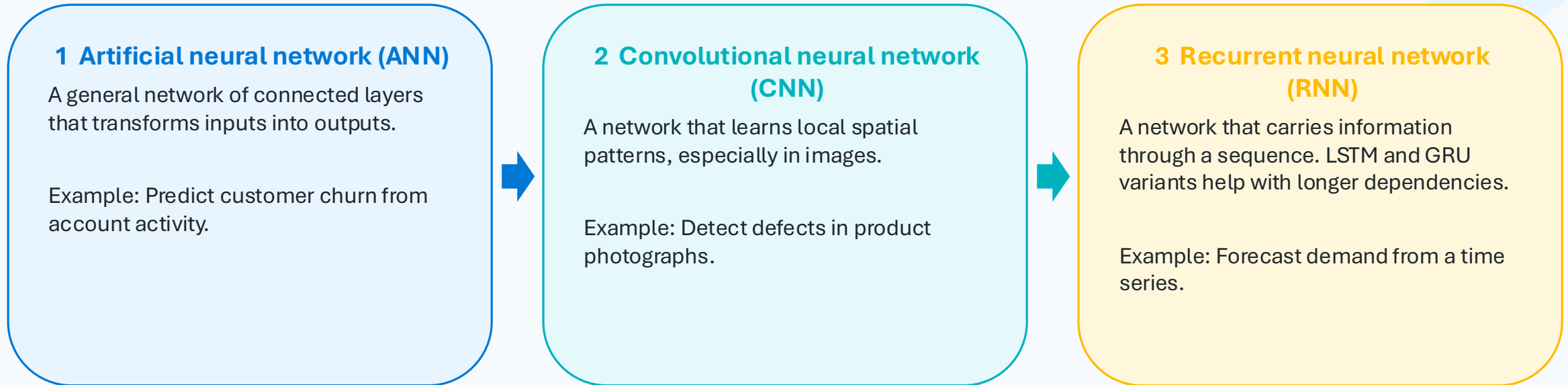
A vision model learns edges, shapes and objects so a factory camera can identify damaged products.

Note

Deep learning powers modern vision, speech, language and multimodal systems.

22 / NEURAL NETWORKS

ANN, CNN and RNN

**Framing question**

Are we predicting a category, estimating a number or discovering groups?

23 / LEARNING TASKS

Classification, regression and clustering

1 Classification

Choose a category from a defined set.

Example: Spam or legitimate email.



2 Regression

Predict a continuous number.

Example: Expected delivery time in minutes.



3 Clustering

Group similar records without predefined labels.

Example: Discover customer behaviour segments.

Framing question

Are we predicting a category, estimating a number or discovering groups?

24 / MODEL TRAINING

Model training

Definition

Training is the process of adjusting model parameters so the model makes fewer errors on examples.

**Mechanism**

The model predicts, a loss function measures error, and an optimiser changes the weights. This repeats across batches and epochs.

**Example**

A photo classifier reviews labelled images many times until it separates healthy and diseased plant leaves reliably.

Note

Training differs from inference, which uses the finished model without repeatedly updating its weights.

25 / MODEL PARAMETERS

Parameters, weights and hyperparameters

Definition

Parameters are values the model learns. Weights are the most common parameters. Hyperparameters are settings chosen by people before or during training.



Mechanism

Training changes weights. Teams choose settings such as learning rate, batch size, architecture and number of training epochs.



Example

A larger learning rate can train faster but may skip over a good solution. A smaller rate can be steadier but slower.

Note

Parameter count affects memory, compute and sometimes capability, but a larger model does not guarantee a better product.

26 / MODEL INFERENCE

Inference and inferencing

Definition

Inference means running a trained model on new input to obtain a prediction, score or generated response. Inferencing refers to the same operational process.

**Mechanism**

The application converts input into the expected representation, sends it to the model and interprets the output.

**Example**

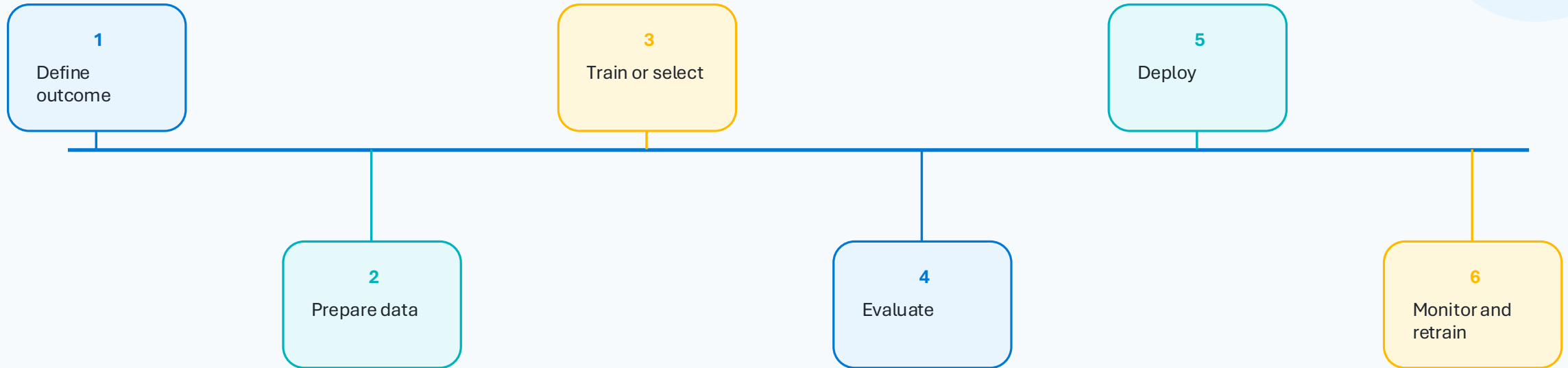
A speech model transcribes a caller while the call is happening. The model performs inference for each audio segment.

Note

Inference design focuses on latency, throughput, cost, availability and privacy.

27 / MODEL WORKFLOW

The end-to-end model workflow

**Key message**

Every model needs traceable data, versioned artefacts, evaluation evidence and an operating owner.

Overfitting and underfitting

Definition

Overfitting happens when a model memorises training details and performs poorly on new data. Underfitting happens when the model is too simple or insufficiently trained.



Mechanism

Teams compare training and validation results, simplify or regularise the model, improve data and tune the learning process.



Example

A fraud model that recognises only last month's exact fraud patterns may miss a new attack and has overfit.

Note

A separate test set estimates generalisation after model selection is complete.

Fine-tuning

Definition

Fine-tuning continues training a pretrained model on carefully selected examples so its behaviour fits a narrower task or style.



Mechanism

The team prepares representative input-output examples, trains cautiously, evaluates regressions and versions the resulting model.



Example

A support organisation fine-tunes a model to produce responses that follow its escalation policy and approved tone.

Note

Use retrieval for changing knowledge. Use fine-tuning when behaviour, format or task performance needs to change.

30 / MODEL EVALUATION

Model evaluation

Definition

Evaluation measures whether a model and the complete application meet quality, safety and operational requirements.

**Mechanism**

Teams use held-out datasets, scenario tests, human review, safety checks and production monitoring. One score rarely describes the whole system.

**Example**

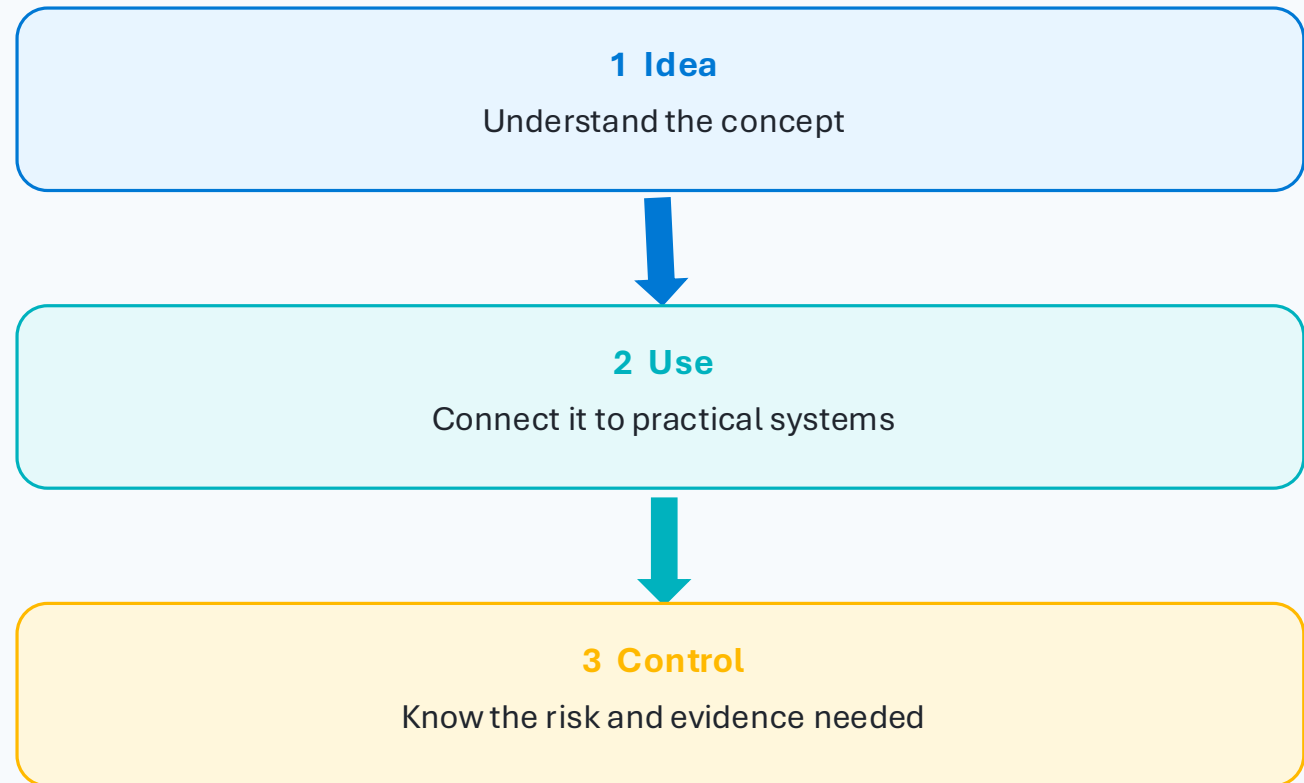
A medical summariser is tested for factual accuracy, omitted information, privacy, latency and performance across patient groups.

Note

Evaluate the model, retrieval, prompts, tools and final user workflow separately and together.

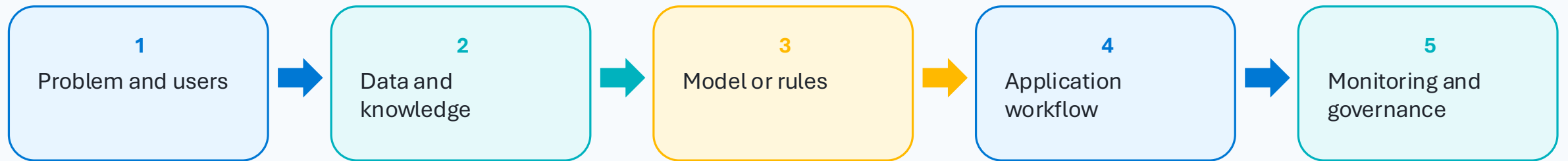
How AI systems work

A useful system connects data, models, software, people and operating controls



32 / SYSTEM ARCHITECTURE

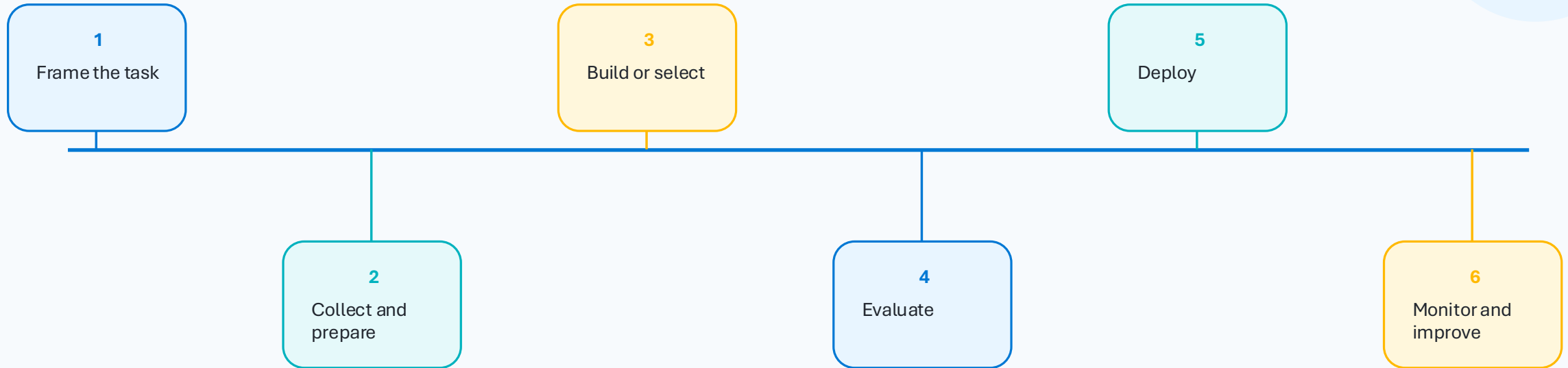
An AI system is more than a model

**Key message**

A strong model cannot rescue a poorly defined problem or unsafe workflow.

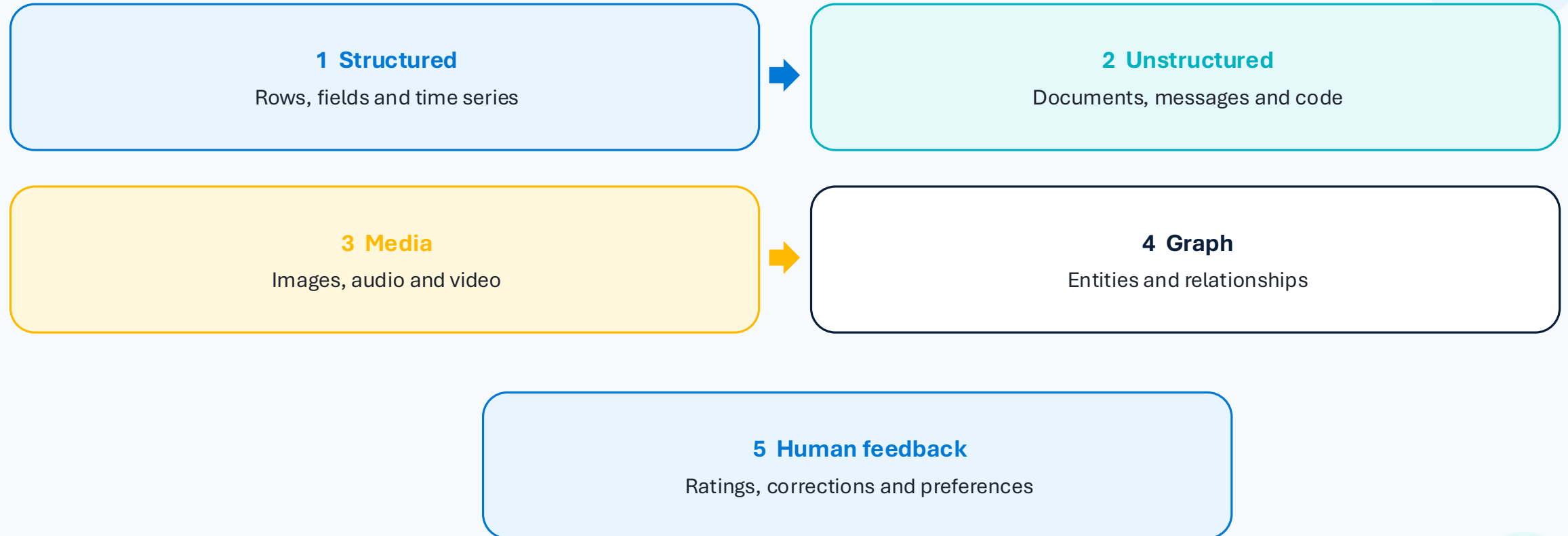
33 / LIFECYCLE

AI development is an iterative operating cycle



34 / DATATYPES

Different tasks require different forms of data



35 / DATA QUALITY

Data teaches both useful patterns and unwanted shortcuts

A Quality questions

- Is the sample representative?
- Are labels consistent?
- Are important cases missing?
- Does the data remain current?
- Can the organisation use it legally?



B Common risks

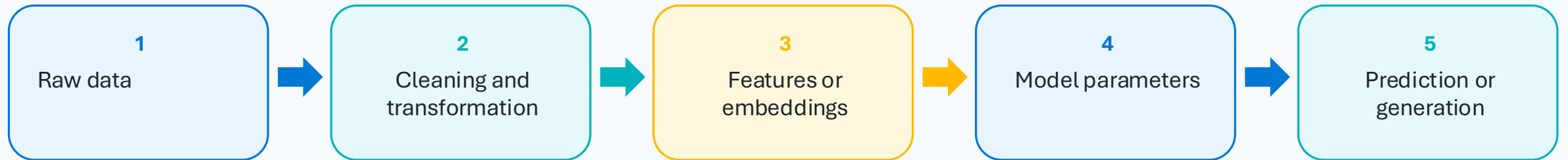
- Bias and historical inequity
- Duplicates and leakage
- Noisy or incorrect labels
- Sensitive information exposure
- Changing conditions after deployment

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

36 / REPRESENTATION

Models need numerical representations

**Key message**

Embeddings place semantically related items near one another in a numerical space.

37 / TRAINING AND INFERENCE

Learning and use are separate phases

A Training

- Uses historical or generated examples
- Adjusts model parameters
- Requires repeated optimisation
- Consumes significant compute for large models



B Inference

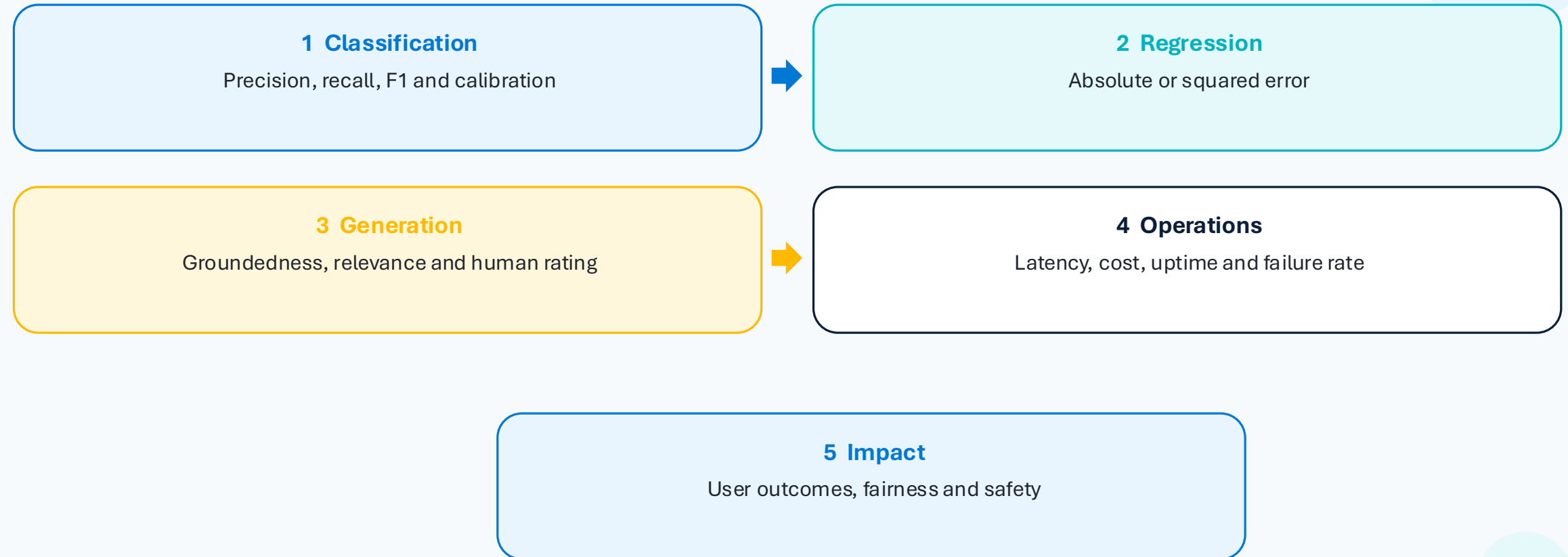
- Receives new inputs
- Runs the learned model
- Returns predictions or generated output
- Must meet latency, cost and reliability needs

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

38 / EVALUATION

Metrics must match the task and the harm



39 / GENERALISATION

Good test results do not guarantee field performance

A During development

- Separate training, validation and test data
- Avoid leakage between datasets
- Test rare and difficult cases
- Compare against a simple baseline



B After release

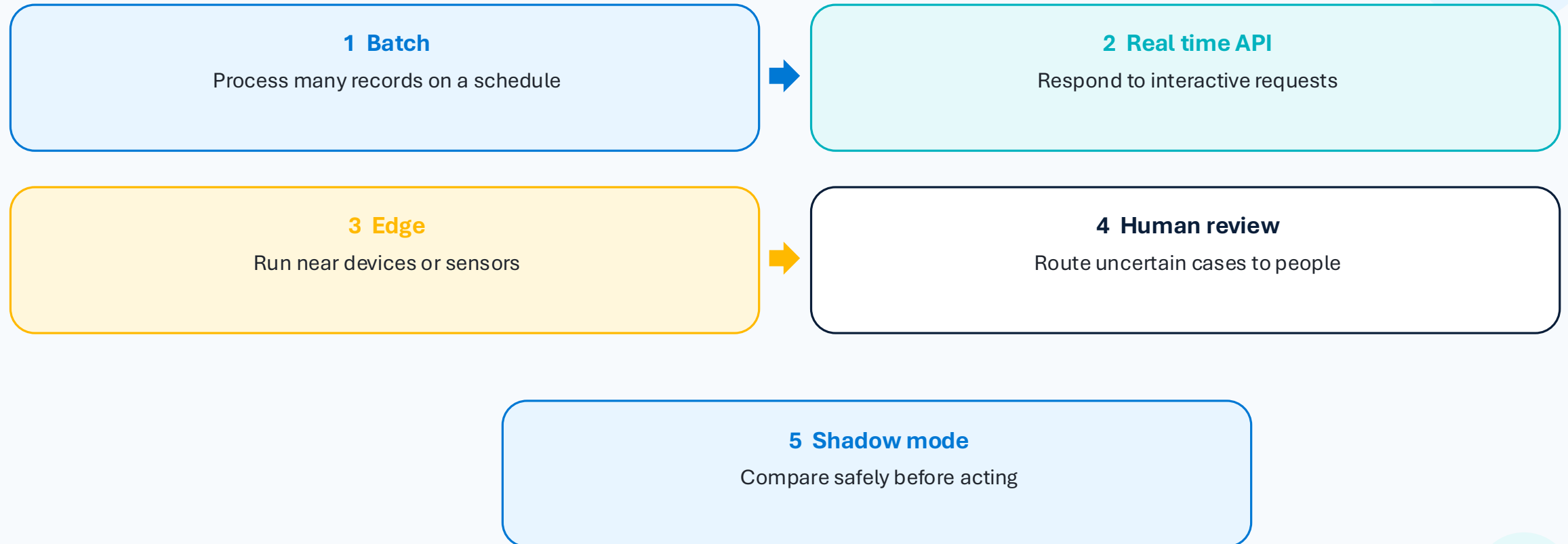
- Watch data and concept drift
- Measure performance by user group
- Capture failures and corrections
- Re-evaluate after model or data changes

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

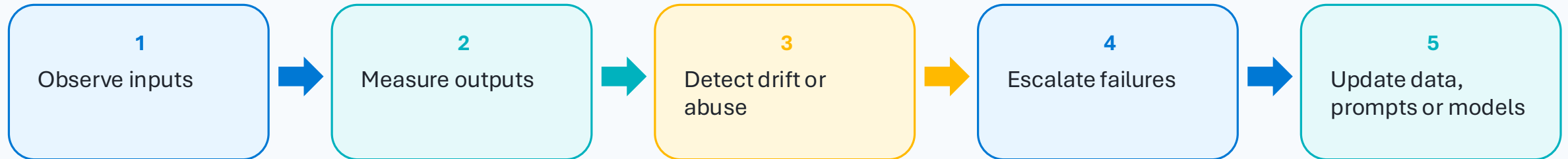
40 / DEPLOYMENT

Deployment patterns reflect risk and workload



41 / OPERATIONS

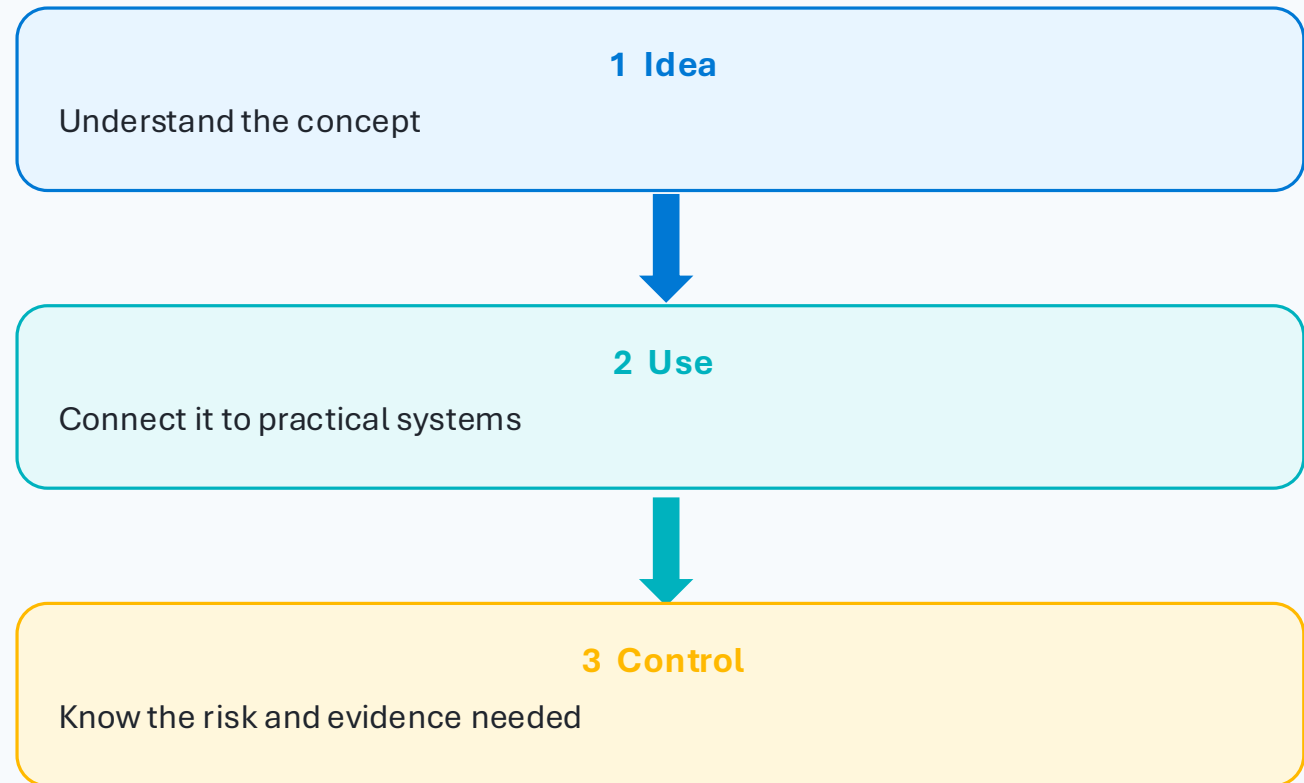
Monitoring closes the learning loop

**Key message**

Production AI needs versioning, logs, evaluation sets, rollback and accountable owners.

Modern generative AI

Foundation models add flexible
language, media and tool-use
capabilities



Deep learning learns layered representations

A Core idea

- Units combine weighted signals
- Layers build increasingly abstract features
- Training changes weights to reduce error
- Scale improves capability when data and compute support it



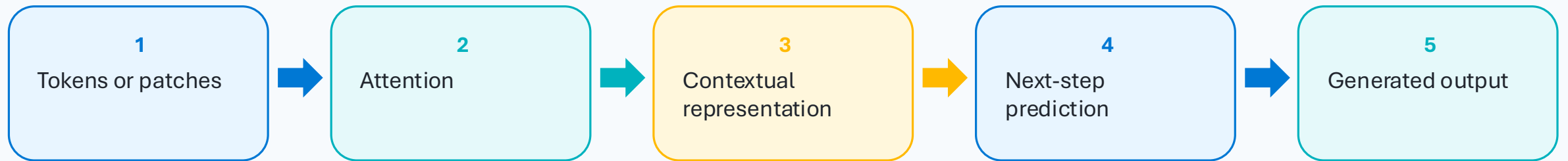
B Practical implications

- Results depend on data and objective design
- Large models require specialised hardware
- Internal reasoning remains difficult to interpret
- Evaluation must cover more than average accuracy

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

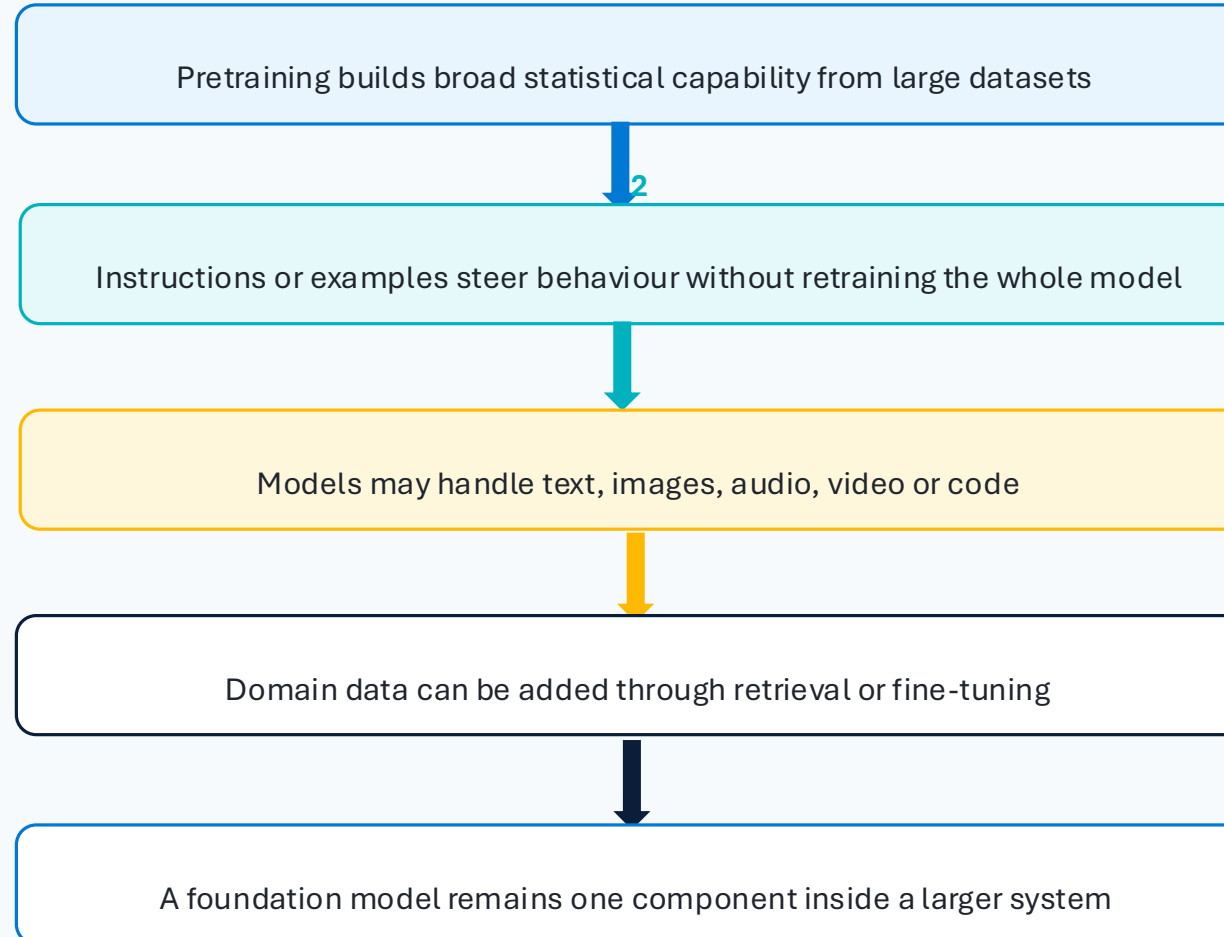
Attention changed how models process sequences



Key message

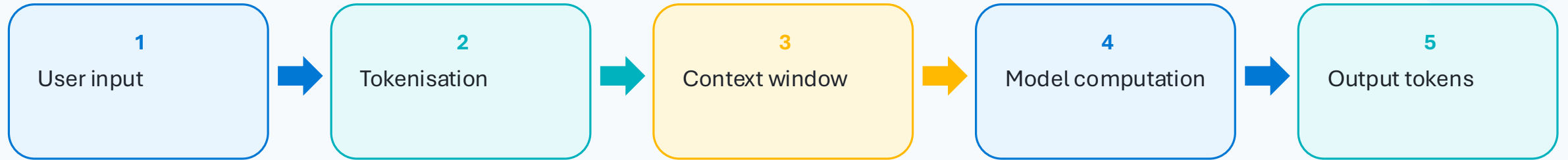
Attention lets a model weigh relationships across the available context.

One pretrained model can support many tasks



46 / TOKENS AND CONTEXT

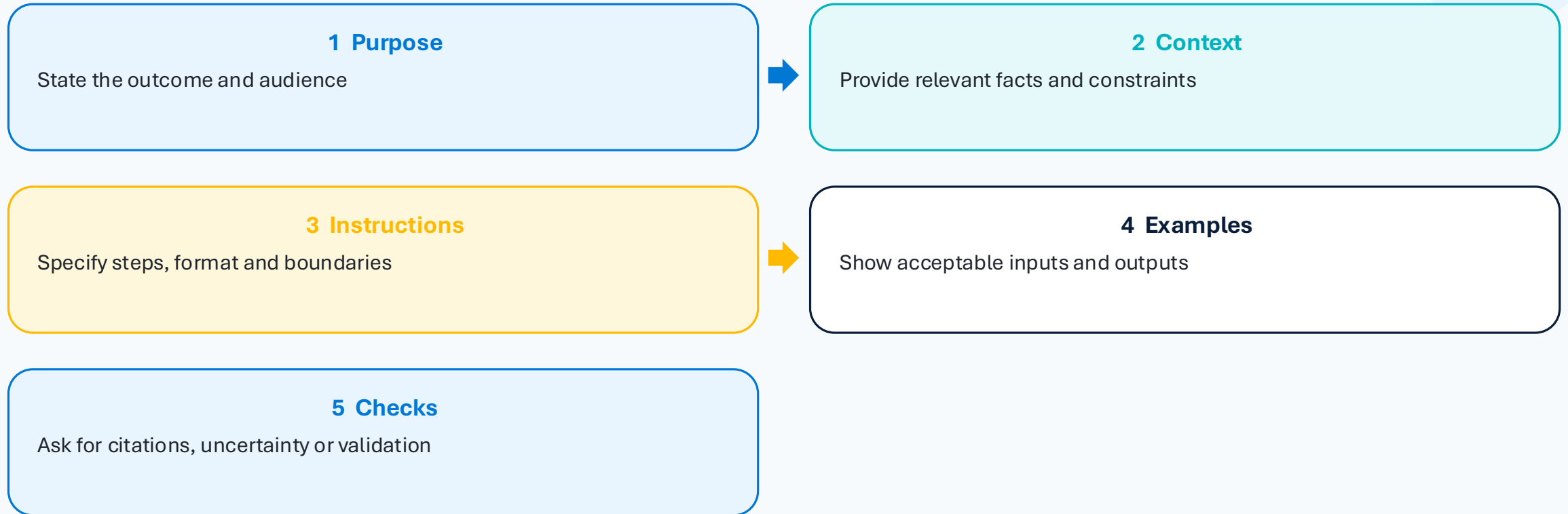
Models operate on tokens inside a limited context window

**Key message**

Longer context increases available information, but relevance and placement still affect results.

47 / PROMPT DESIGN

A strong prompt makes the task testable



48 / GENERATION BEHAVIOUR

Fluent output can still be wrong

A Why errors occur

- The model predicts plausible sequences
- Training data may be incomplete or conflicting
- Prompts may omit necessary context
- Tools or retrieval may return weak evidence



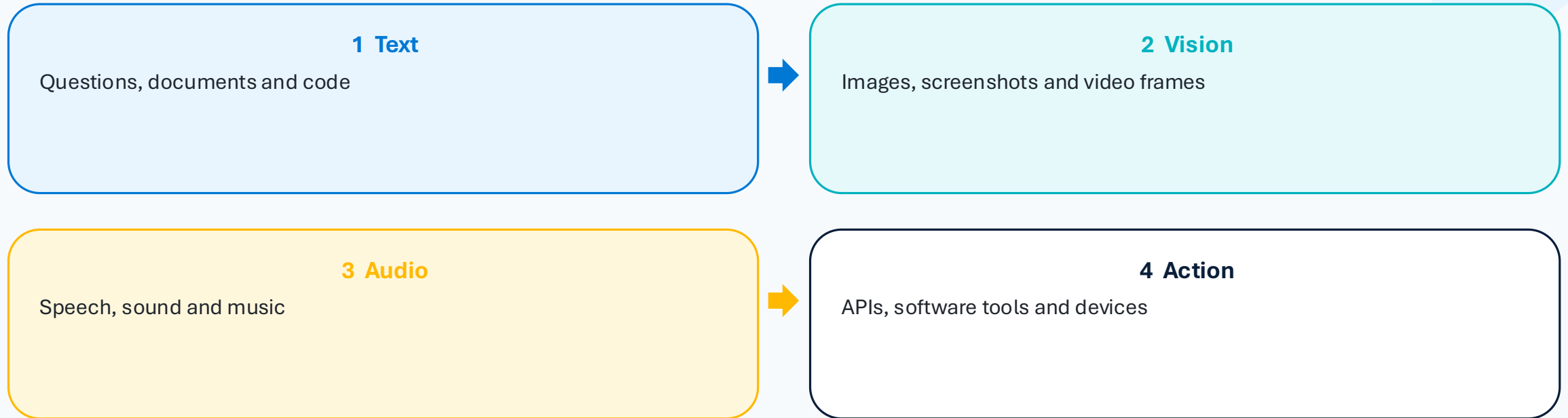
B How teams reduce risk

- Ground answers in trusted sources
- Use structured output and validation
- Evaluate with representative test cases
- Require human review for consequential decisions

? Decision lens

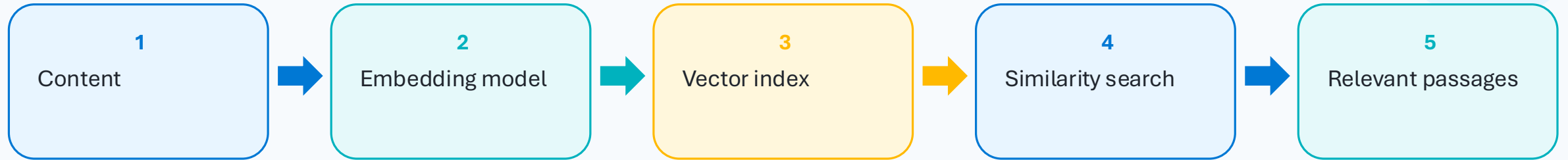
Choose based on the problem, the data type, explainability needs and the level of ambiguity.

Multimodal systems connect more than one information type



50 / EMBEDDINGS

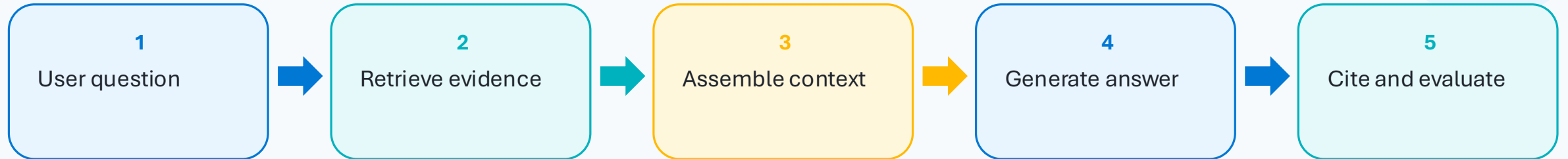
Vector search retrieves by meaning

**Key message**

Semantic retrieval complements keywords, filters and access controls.

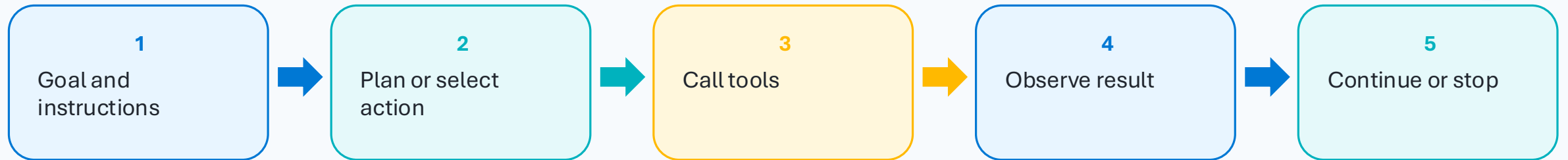
51 / RETRIEVAL-AUGMENTED GENERATION

RAG grounds generation in selected knowledge

**Key message**

RAG improves access to current or private knowledge, but poor retrieval still produces poor answers.

Agents combine models with goals, tools and state

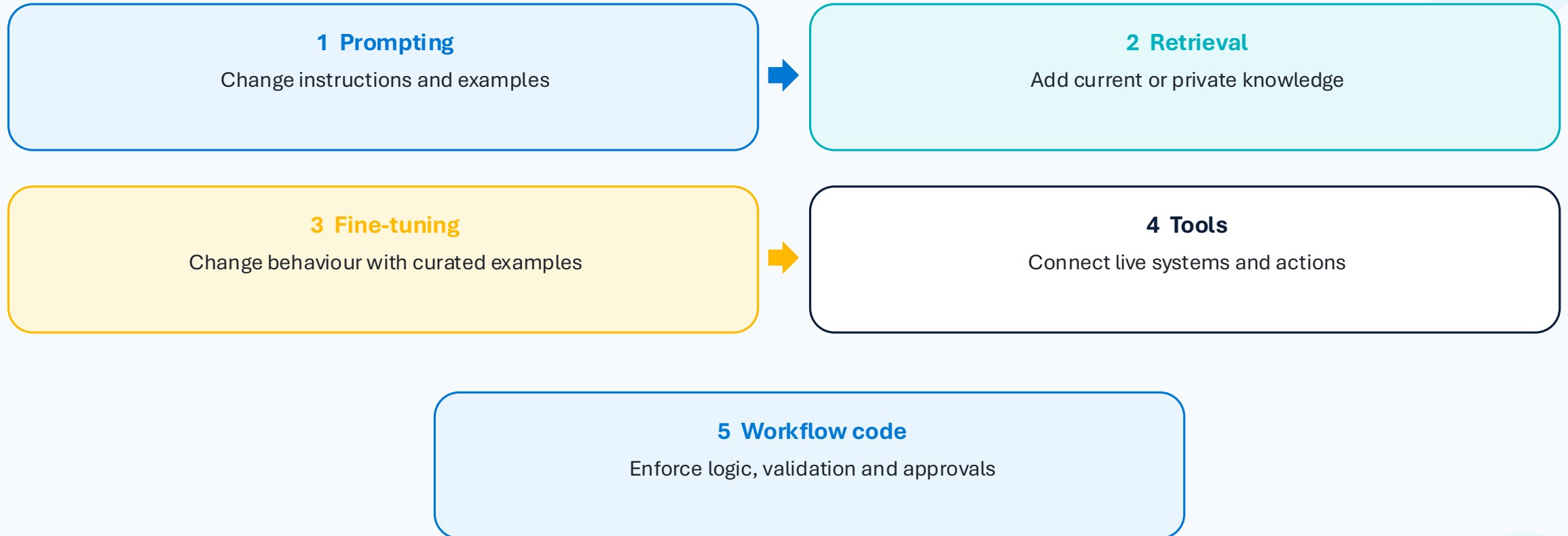


Key message

Use an agent only when the task needs flexible multi-step decisions. Deterministic workflows remain safer for fixed processes.

53 / CUSTOMISATION

Choose the lightest method that solves the gap



Open and proprietary models involve different trade-offs

A Open-weight models

- Greater hosting and modification control
- Potential on-premises or edge deployment
- Team owns more operational complexity
- Licences and model cards require review



B Managed proprietary models

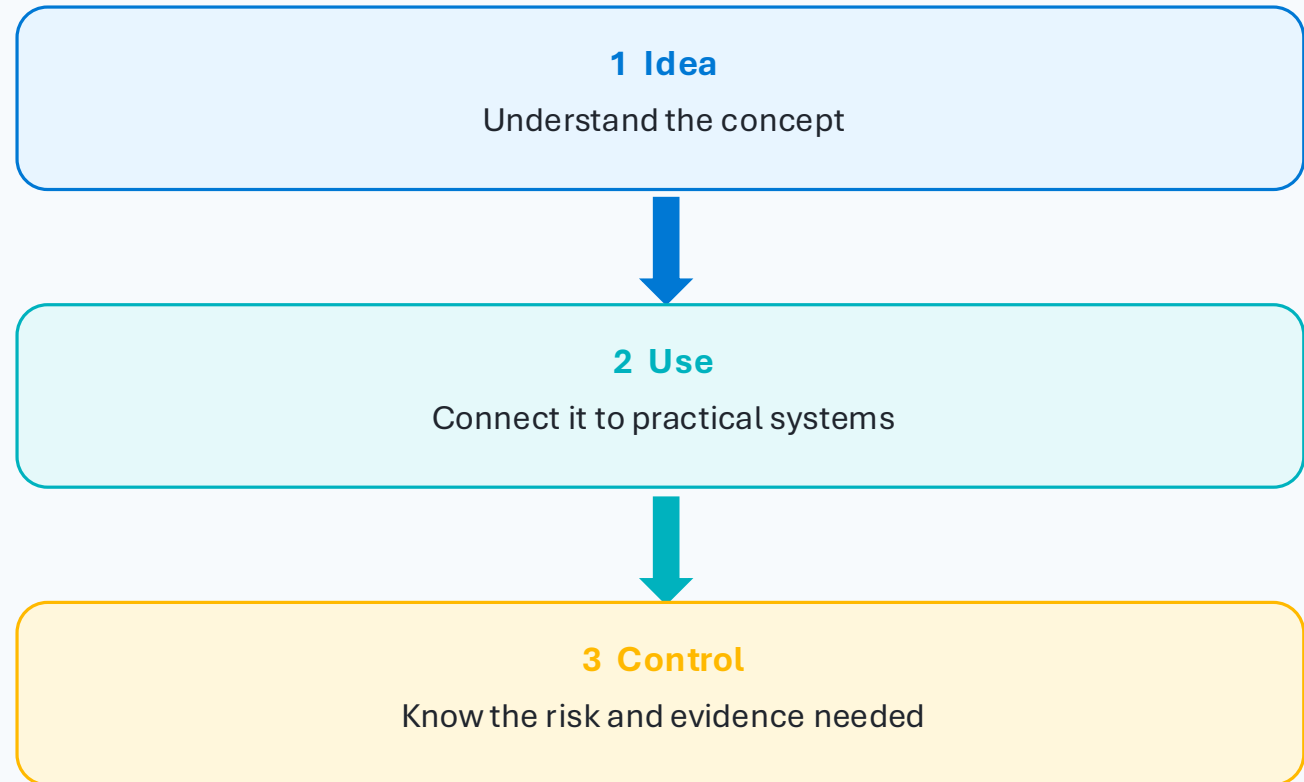
- Faster access through hosted APIs
- Provider manages core infrastructure
- Less control over model internals
- Cost, residency and provider terms matter

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

Language models and agents

Tokens, context, tools, memory and protocols turn models into working systems



Generative AI

Definition

Generative AI creates new text, code, images, audio, video or structured data from patterns learned during training.



Mechanism

A generative model estimates a plausible next element, denoises a sample or otherwise constructs output from a learned distribution.



Example

A product team turns a written campaign brief into draft copy, images and regional variations for human review.

Note

Generative AI can create useful drafts, but fluent output may be inaccurate, unsafe or legally restricted.

57 / LARGE LANGUAGE MODEL

Large language model (LLM)

Definition

An LLM is a large neural language model trained on extensive text and code so it can understand and generate sequences of tokens.



Mechanism

Most current LLMs use transformer architectures. The model predicts output tokens using the prompt and information inside its context window.



Example

An LLM summarises a policy document, answers questions and drafts code from natural-language instructions.

Note

LLMs include general, reasoning, code and domain-focused models. A small language model may reduce cost and latency for narrower work.

Large and small language models

A Large language models

- Broader capability across tasks
- Often stronger reasoning and instruction following
- Higher inference cost and latency
- Usually require cloud-scale infrastructure



B Small language models

- Lower cost and faster responses
- May run on local or edge hardware
- Easier to specialise for narrow tasks
- Need careful testing outside their focus

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

59 / TOKEN BASICS

Token, tokenisation and context window

1 Token

A unit the model reads or writes. It may be a word, part of a word, punctuation or another encoded element.

Example: “unbelievable” may become several tokens.



2 Tokenisation

The process that converts input into token IDs and converts generated IDs back into readable output.

Example: A tokenizer prepares a prompt before model inference.



3 Context window

The maximum token space available for instructions, conversation, retrieved evidence and output.

Example: A long contract may need chunking or retrieval.

Framing question

Are we predicting a category, estimating a number or discovering groups?

Temperature and sampling

Definition

Temperature adjusts how strongly a generative model favours its highest-probability next token. Lower values usually make output more consistent. Higher values increase variety.



Mechanism

The model produces token probabilities. Sampling settings reshape or limit the choices before selecting the next token.



Example

Use low temperature for extracting invoice fields. Use moderate variation when brainstorming campaign headlines.

Note

Temperature does not improve factual accuracy. Grounding, validation and evaluation remain necessary.

61 / PROMPTING

Prompting and prompt engineering

A Prompting

- Giving a model an instruction or question
- Adding context or examples
- Useful for one-off or exploratory work
- Can be informal when risk is low



B Prompt engineering

- Designing repeatable prompt structures
- Specifying inputs, outputs and constraints
- Testing variants against an evaluation set
- Versioning prompts like application code

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

62 / MULTIMODAL MODELS

Multimodal model

Definition

A multimodal model accepts or produces more than one data type, such as text, images, audio or video.

**Mechanism**

The system encodes different modalities into representations the model can relate within one task.

**Example**

A maintenance assistant examines a machine photo, reads the technician's note and explains the likely fault.

Note

Multimodal systems need separate tests for each modality and for cross-modal misunderstandings.

Knowledge and grounding

D Definition

Knowledge is the information an AI system can use to answer or act. Grounding connects an output to supplied, current or authoritative evidence.



M Mechanism

A system can use information learned during training, content placed in the prompt, retrieved documents or live tool results.



E Example

A benefits assistant retrieves the current employee handbook and cites the exact section used for its answer.

Note

Model training knowledge can become outdated. Retrieval and tools add current or private information.

Agent memory

D Definition

Memory is stored information an agent can reuse across steps or conversations. It may include recent messages, user preferences, task state or summaries.



M Mechanism

Short-term memory stays in the active context. Long-term memory uses an external store and a retrieval policy.



E Example

A travel agent remembers the approved budget and chosen dates while comparing several itineraries.

Note

Memory requires consent, retention rules, access controls and a way to correct or delete stored information.

Tools, tool calls and external systems

1 Tool

A capability the model can request, such as search, calculation, code execution or an API.

Example: A weather lookup function.



2 Tool call

A structured request containing the selected tool and arguments. Application code executes it.

Example: `get_weather(city="Valletta")`.



3 External system

The database, SaaS platform, device or service that performs the real operation.

Example: A CRM creates the customer case.

AI agent

Definition

An AI agent uses a model to interpret a goal, choose steps, call tools, observe results and continue until it finishes or needs help.



Mechanism

Instructions define behaviour. Tools enable action. State records progress. Guardrails and approvals limit what the agent may do.



Example

An IT support agent checks device status, searches known issues and opens a ticket when automated repair fails.

Note

A fixed workflow is preferable when every step can be known in advance.

Single-agent and multi-agent systems

A Single agent

- One agent coordinates the whole task
- Simpler state, evaluation and debugging
- Lower communication overhead
- Best for bounded workflows



B Multi-agent system

- Specialist agents divide complex work
- Agents may review or delegate to one another
- Requires coordination and shared task state
- Creates more latency, cost and failure paths

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

Agent Skills

Definition

An Agent Skill is a reusable package of instructions and optional resources that teaches an agent how to perform a defined task.



Mechanism

The agent first sees the skill name and description. It loads the full instructions and supporting resources only when the task matches.



Example

An expense-report skill explains company limits, validation steps and the required reimbursement template.

Note

Skills reduce repeated prompt text and allow teams to version domain guidance independently from agent code.

SKILL.md, .agent.md and AGENTS.md

Reusable capability

- SKILL.md defines one Agent Skill
- YAML metadata describes when it applies
- The body contains task instructions
- References, scripts and assets may sit beside it



Agent and repository instructions

- .agent.md defines a custom VS Code agent profile
- It can select instructions, tools and a model
- AGENTS.md gives coding agents repository guidance
- Support varies by agent host

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

Model Context Protocol (MCP)

Definition

MCP is an open protocol that lets an AI application discover and use tools, resources and prompts exposed by external servers.



Mechanism

A host creates an MCP client connection. The client lists server capabilities and sends structured requests when the agent chooses one.



Example

A coding assistant connects to a GitHub MCP server to read issues and create a pull request with authorised credentials.

Note

MCP connects an agent to tools and data. A2A connects one agent application to another.

71 / MCP ARCHITECTURE

MCP client and MCP server

A MCP client

- Runs inside the agent host or application
- Connects to one MCP server
- Discovers tools and resources
- Sends requests and receives results



B MCP server

- Publishes a defined capability surface
- Wraps APIs, files, databases or services
- Enforces authentication and validation
- Returns structured content or tool results

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

Agent2Agent (A2A) protocol

D Definition

A2A is an open protocol for communication and collaboration between independent agent applications.



M Mechanism

An agent publishes an Agent Card describing its capabilities. Another agent can discover it, send tasks and receive messages, files or structured results.



E Example

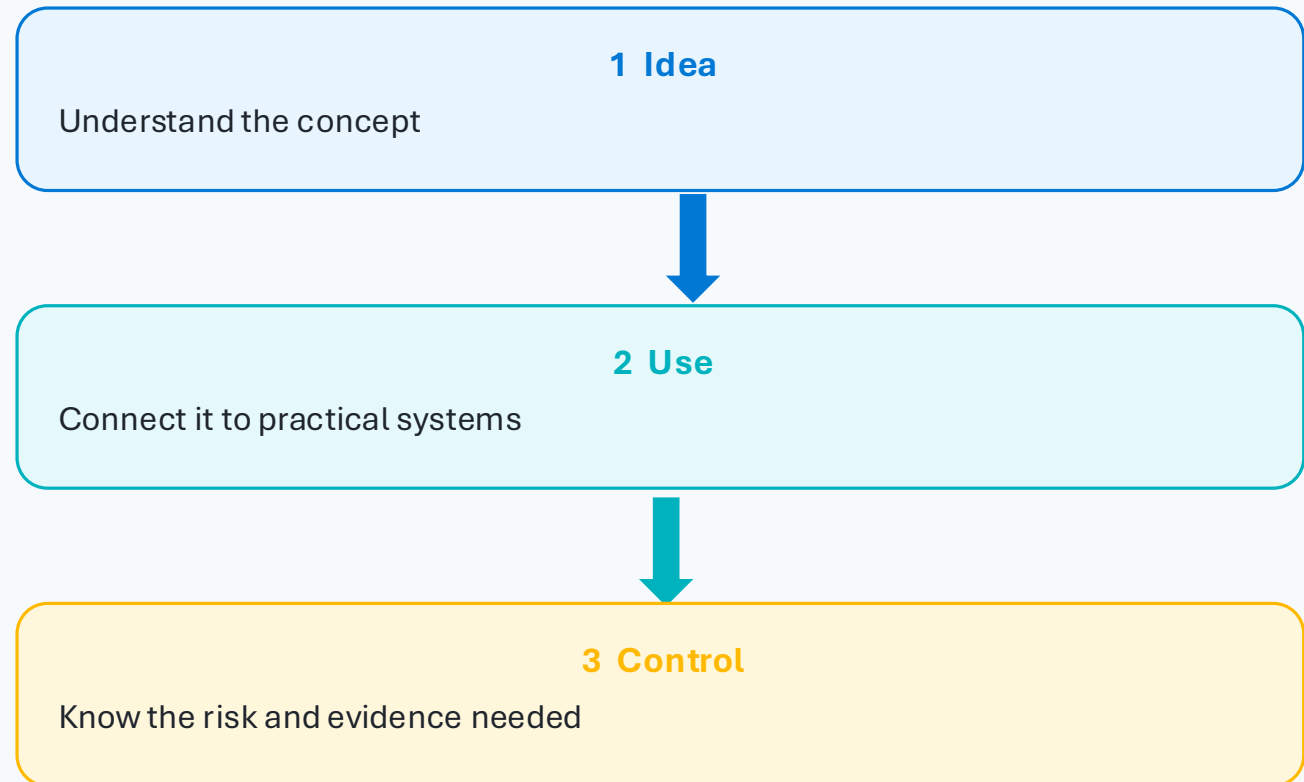
A customer-service agent delegates a refund investigation to a finance agent built by another team.

Note

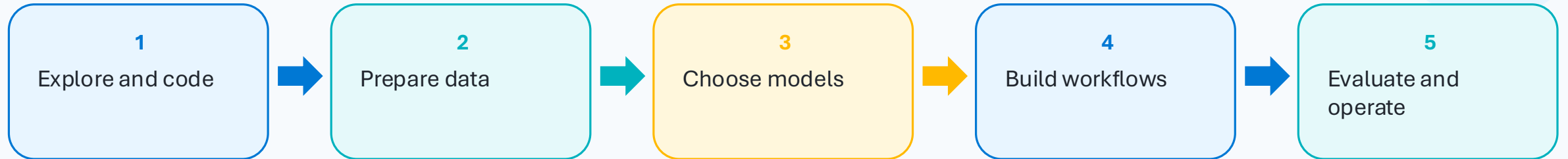
A2A supports agent interoperability without exposing each agent's internal memory, model or tools.

The AI tool ecosystem

Tools support different stages, roles and levels of control



A complete solution uses several tool layers



Key message

Tool choice should follow the problem, team skills, risk and deployment environment.

Notebooks and IDEs support different working styles

A Notebooks

- Jupyter and hosted notebook services
- Good for exploration and teaching
- Combine code, output and explanation
- Require discipline for reproducibility



B IDEs

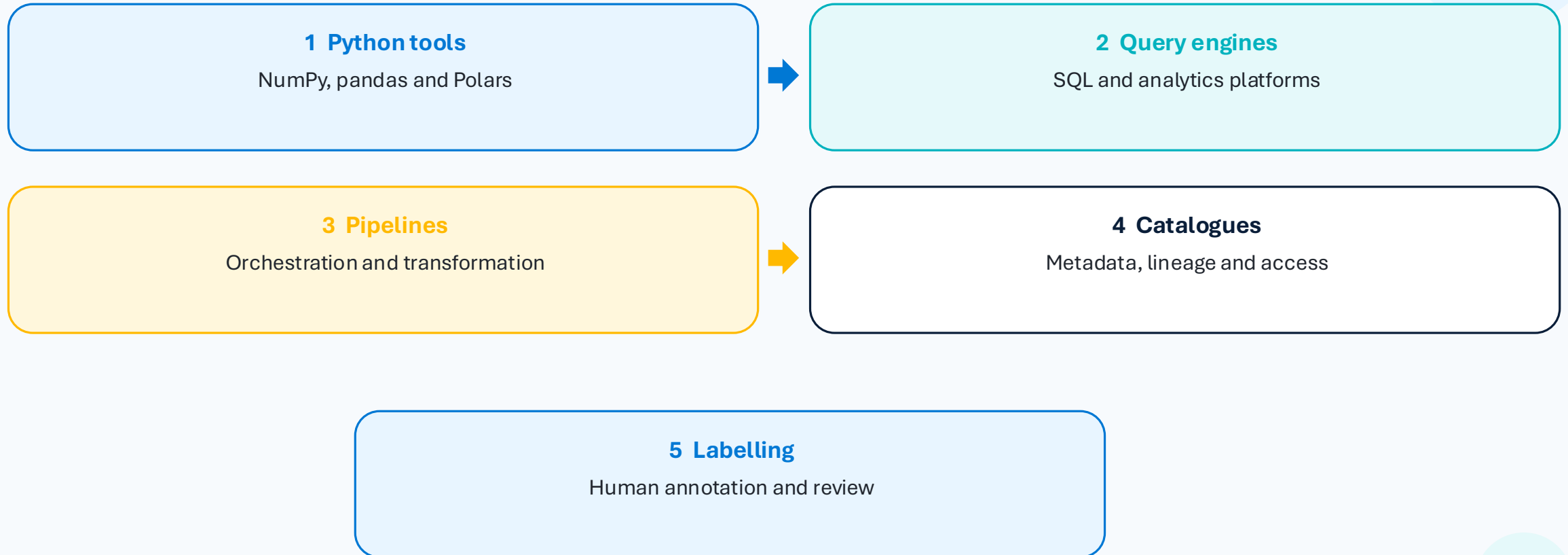
- Visual Studio Code and similar editors
- Good for structured applications
- Support source control, testing and debugging
- Fit team development and deployment workflows

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

76 / DATATOOLS

Data work spans preparation, storage and governance



Frameworks cover classical and deep learning

A Classical machine learning

- scikit-learn
- XGBoost and LightGBM
- Statistical libraries
- Strong baselines for structured data



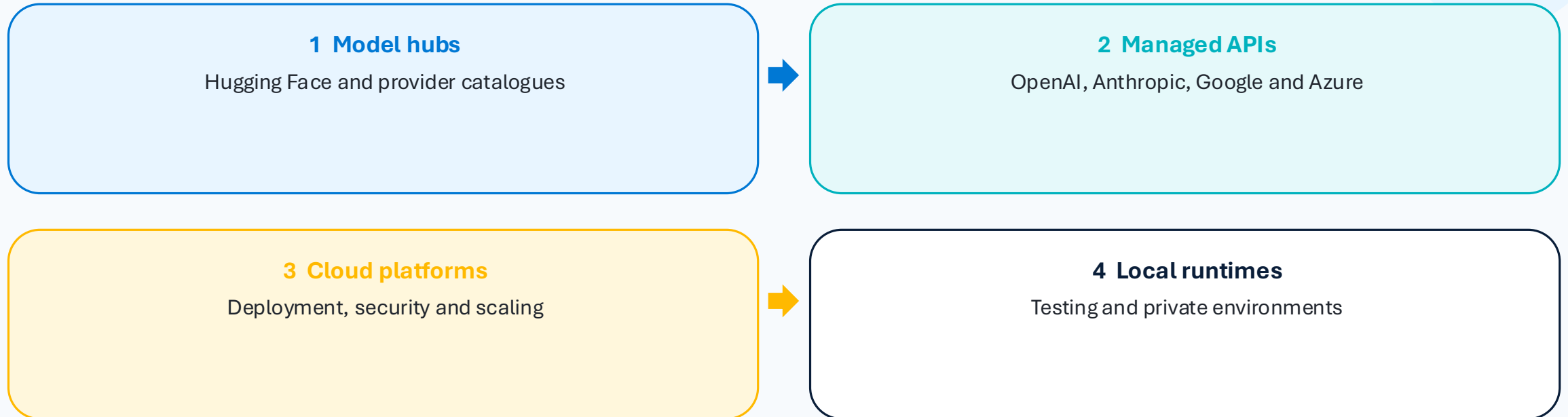
B Deep learning

- PyTorch
- TensorFlow and Keras
- JAX
- GPU and accelerator libraries

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

Teams can build, download or call models



79 / ORCHESTRATION

Frameworks help connect models, tools and state

A Useful capabilities

- Prompt and workflow composition
- Tool calling and structured output
- Memory or state management
- Tracing and evaluation hooks



B Questions before adopting

- Can simple code solve the task?
- How will versions be tested?
- Can failures be inspected?
- What security boundary does the framework create?

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

Retrieval systems need more than a vector database

1

Chunking determines what information the system can retrieve

2

Metadata and filters enforce scope and permissions

3

Hybrid search combines semantic and keyword matching

4

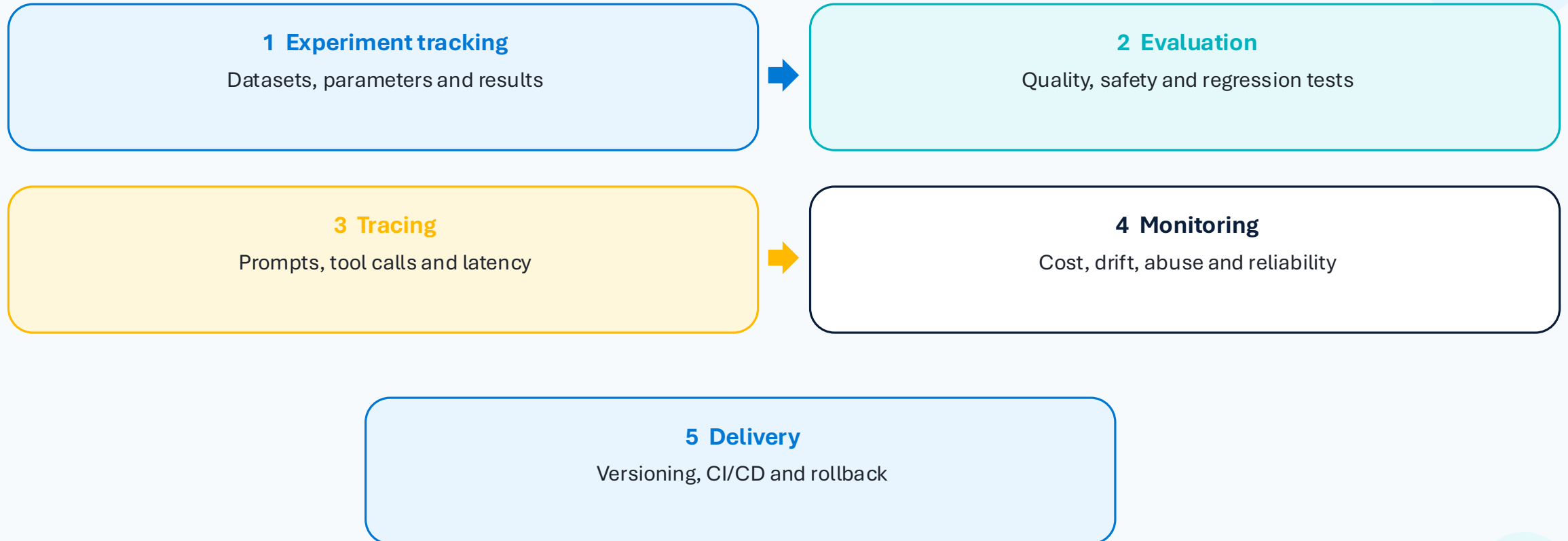
Reranking improves the order of candidates

5

Evaluation measures retrieval quality before judging generation

81 / EVALUATION AND OPERATIONS

Production tooling makes behaviour inspectable



82 / BUILDING WITHOUT DEEP CODE

No-code and copilots widen access

A No-code and low-code

- Visual workflows and connectors
- Useful for bounded business processes
- Faster prototyping for domain experts
- Still requires permissions and testing



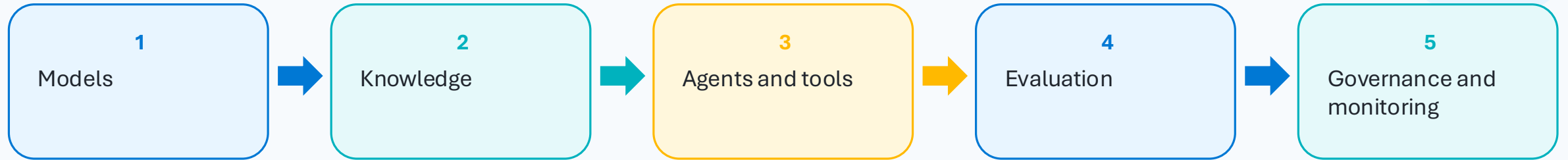
B Copilots and assistants

- Writing, research and software support
- Help users work inside familiar tools
- Require verification and data controls
- Do not remove professional accountability

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

Foundry connects models, agents, knowledge and operations

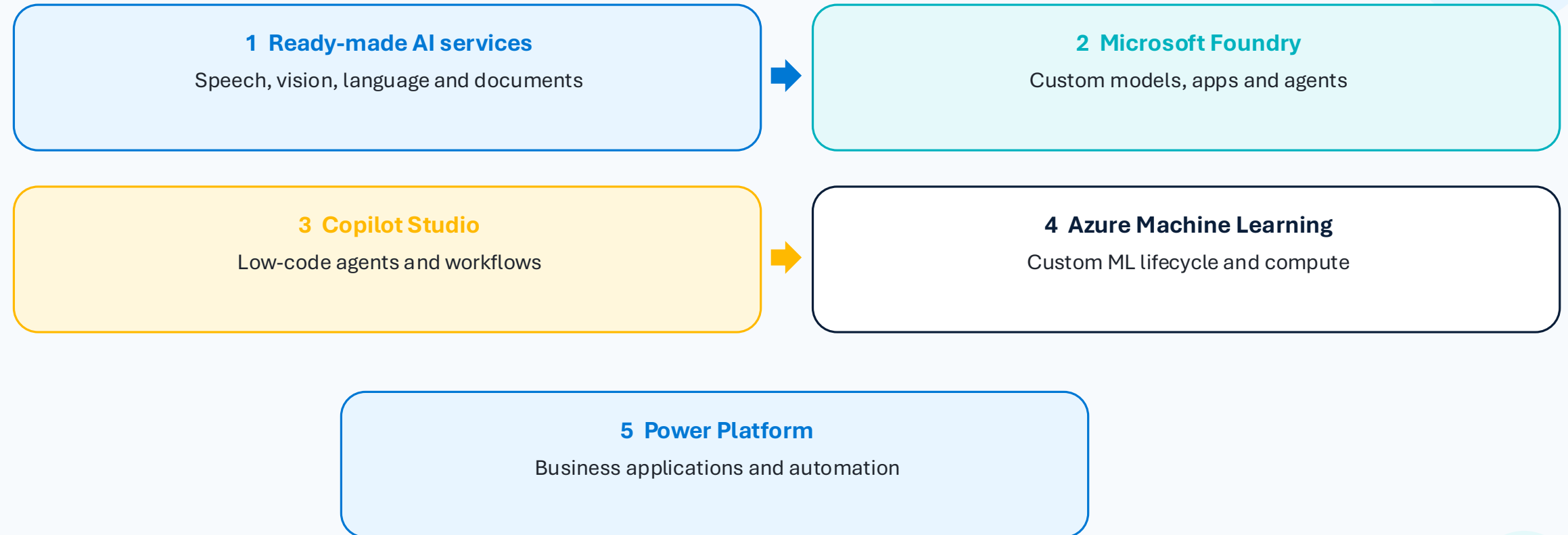


Key message

Foundry provides a managed environment for building, optimising and governing AI applications and agents.

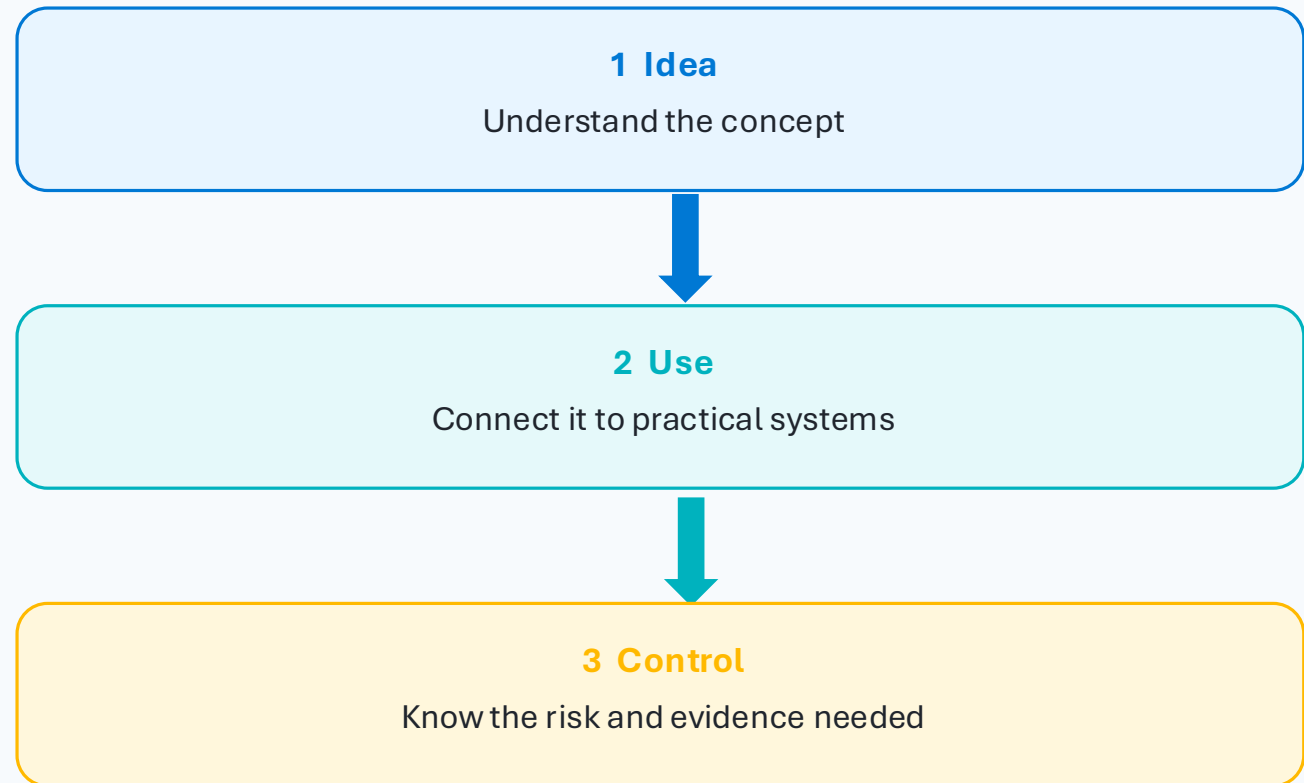
84 / MICROSOFT SERVICE CHOICE

Choose the level of control the solution requires



Microsoft AI platform concepts

Models, knowledge, agents, identity
and specialised services work
together



Microsoft Foundry Models

D Definition

Foundry Models is a catalogue and deployment experience for discovering, comparing, customising and operating models in Azure.



M Mechanism

Teams inspect model cards, compare benchmarks, choose a deployment option and expose the model through a secured endpoint.



E Example

A team compares a small language model with a larger reasoning model before deploying the least expensive option that passes its tests.

Note

The catalogue includes models sold by Azure plus models supplied by partners and the community.

Azure-sold and third-party Foundry Models

A Models sold by Azure

- Microsoft provides the Azure commercial relationship
- Azure integration and enterprise support
- Serverless deployment options may be available
- Review model-specific data and regional terms



B Partner and community models

- Providers include external companies and communities
- Specialised and open-weight options are available
- Provider support and licences vary
- Deployment may use serverless APIs or managed compute

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

Azure AI Search

D Definition

Azure AI Search is a managed information-retrieval service that supports keyword, vector, hybrid and semantic search.



M Mechanism

Content is indexed with text, metadata and optional embeddings. Hybrid search runs keyword and vector retrieval together and merges the ranked results.



E Example

A support copilot searches product names exactly while also finding semantically related troubleshooting passages.

Note

Azure AI Search provides the retrieval foundation used by Foundry IQ knowledge bases.

Foundry IQ

D Definition

Foundry IQ is a managed knowledge layer that gives agents permission-aware access to enterprise information with citations.



M Mechanism

A knowledge base connects one or more sources. Agentic retrieval can plan subqueries, search sources, rerank results and return grounded evidence.



E Example

An HR agent answers policy questions from SharePoint and Azure storage while respecting each employee's document permissions.

Note

Foundry IQ uses Azure AI Search and can serve Foundry agents, Microsoft Agent Framework or custom applications.

Microsoft Foundry Agent Service

D Definition

Foundry Agent Service provides managed infrastructure to create, host, publish, evaluate and monitor AI agents.



M Mechanism

Developers can configure prompt agents or deploy custom hosted agent code, attach models and tools, then expose a stable endpoint.



E Example

A procurement agent checks supplier records, queries policy knowledge and prepares a purchase review for approval.

Note

Enterprise features include identity, networking, observability, content safety and integration with Agent 365.

Microsoft Agent 365

D Definition

Agent 365 is a control plane for observing, securing and governing agents across an organisation.



M Mechanism

Administrators maintain an agent registry, monitor activity and risk, apply access controls and connect Microsoft security and compliance services.



E Example

A security team sees which agents access finance data and investigates an agent that begins making unusual tool calls.

Note

Agent 365 governs agents from several platforms. It does not replace the model or the agent-building framework.

Microsoft Entra ID, authentication and RBAC

D Definition

Microsoft Entra ID proves the identity of a person, application or agent. Azure role-based access control (RBAC) determines which actions that identity may perform at a scope.



M Mechanism

An agent obtains an access token, presents it to the target service and succeeds only when the assigned role permits the requested action.



E Example

A research agent receives read access to one storage container but cannot delete documents or access the payroll container.

Note

Use least privilege, managed or agent identities, scoped roles and audit logs. Authentication answers who; authorisation answers what they may do.

Azure Policy for AI governance

D Definition

Azure Policy evaluates Azure resources against organisational rules and can audit, deny or remediate non-compliant configurations.



M Mechanism

Administrators assign built-in or custom policy definitions at a management group, subscription or resource-group scope.



E Example

A policy blocks Foundry model deployments unless the publisher or model asset appears on an approved list.

Note

Policy complements RBAC. RBAC controls who can act, while policy controls which resource configurations the organisation permits.

Azure Document Intelligence

D Definition

Document Intelligence extracts text, tables, layout and structured fields from documents and images.



M Mechanism

Prebuilt or custom models analyse files and return machine-readable results with confidence and location information.



E Example

An accounts-payable workflow extracts supplier, total, tax and line items from invoices before a person approves payment.

Note

Use Content Understanding when documents require flexible generative extraction across text, images, audio or video.

Speech to text and text to speech

D Definition

Azure Speech converts spoken audio into text and synthesises text as speech. It also supports translation, language identification and voice scenarios.



M Mechanism

Applications stream or upload audio through SDKs or APIs, then receive transcripts, timing information or generated audio.



E Example

A meeting assistant produces live captions and a searchable transcript, while a voice bot reads an approved response to a caller.

Note

Evaluate accents, noise, latency, consent, accessibility and the risks of synthetic voice.

Azure Language and Azure Translator

A Azure Language

- Analyses text with targeted NLP models
- Sentiment and opinion mining
- Named entities and personal-data detection
- Summarisation and question answering



B Azure Translator

- Real-time text translation
- Batch and document translation
- Custom terminology for a domain
- Useful for multilingual applications

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

Computer Vision and Content Understanding

A Azure Vision

- Image analysis and optical character recognition
- Detects visual features and text
- Useful for accessibility and inspection
- Custom Vision supports specialised classifiers



B Azure Content Understanding

- Extracts schema-defined information
- Works across documents, images, audio and video
- Produces grounded, structured output
- Useful when prebuilt document models do not fit

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

Azure Machine Learning and MLOps

D Definition

Azure Machine Learning supports the lifecycle for custom machine-learning models, data, experiments, compute, deployments and monitoring.



M Mechanism

Teams version code and data, track experiments, register models, automate pipelines and promote tested deployments through environments.



E Example

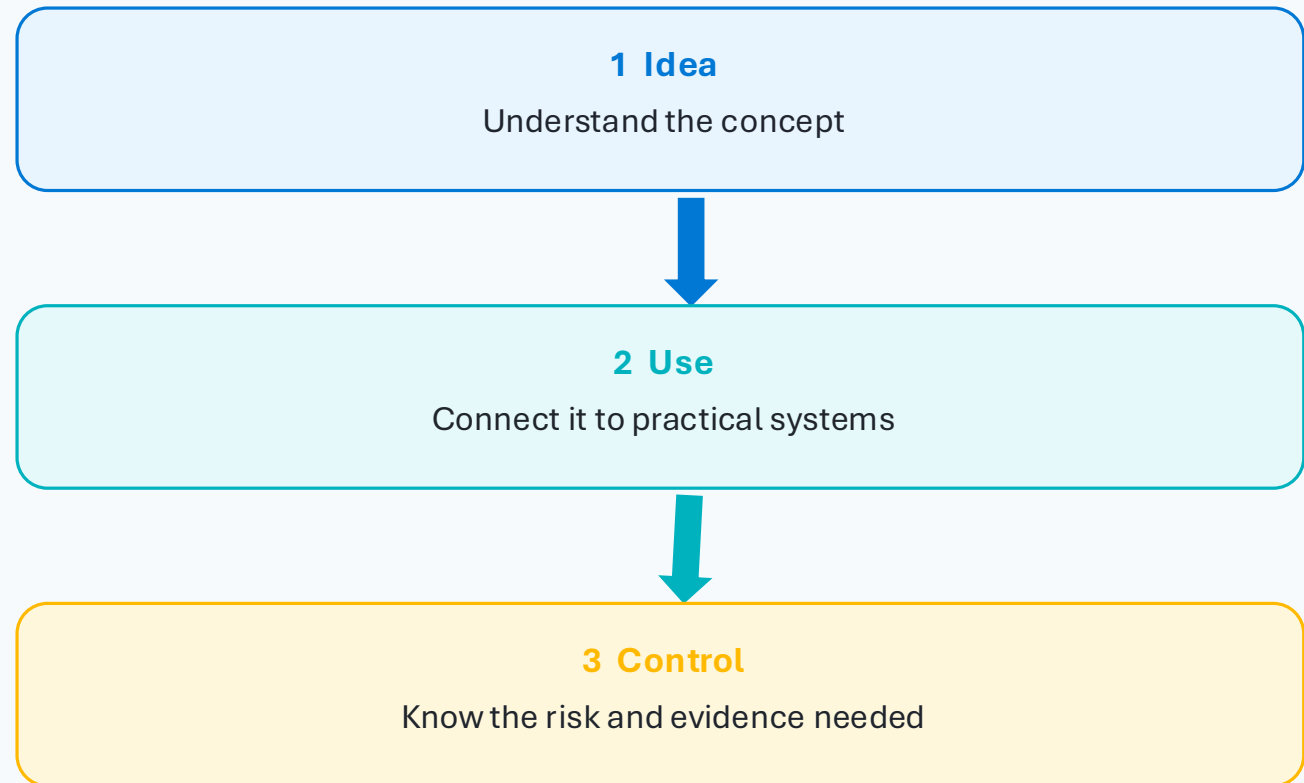
A manufacturer retrain a defect model each month, compares it with the current version and deploys only when quality gates pass.

Note

MLOps applies software delivery and operations practices to data and model changes.

Safe and responsible use

Capability, risk and accountability must be designed together



100 / LIMITATIONS

Every AI system has technical and social limits

A Technical limits

- Errors and fabricated content
- Sensitivity to data and context
- Performance drift over time
- Uncertain behaviour outside tested conditions



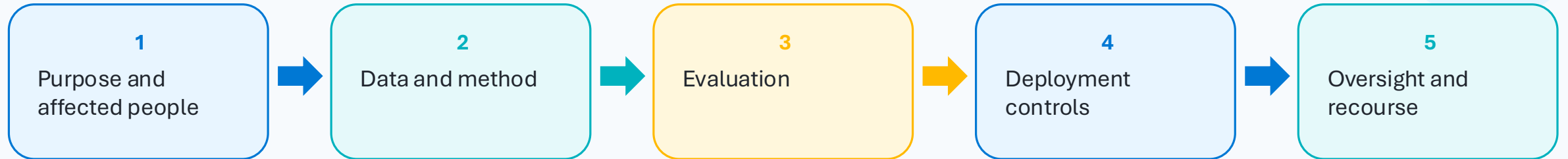
B Human and social limits

- Bias and unequal impact
- Privacy and intellectual property concerns
- Automation bias and overreliance
- Unclear responsibility for harm

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

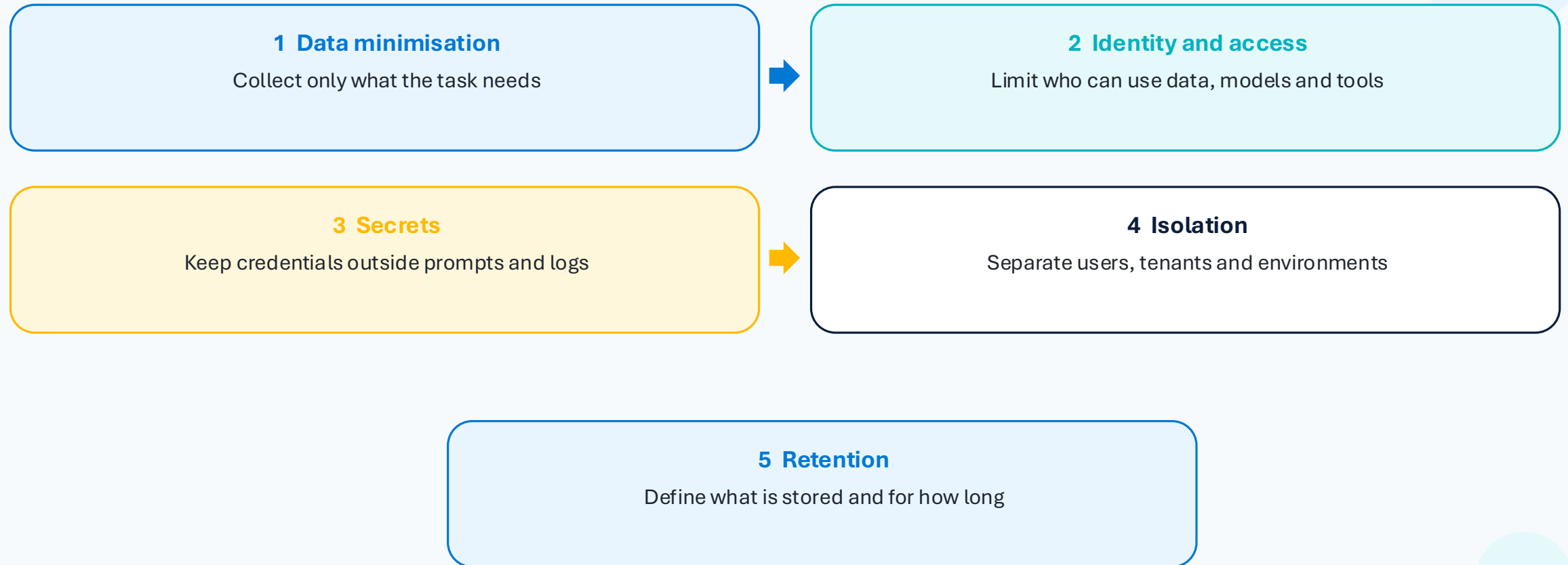
Responsibility begins before a model is selected



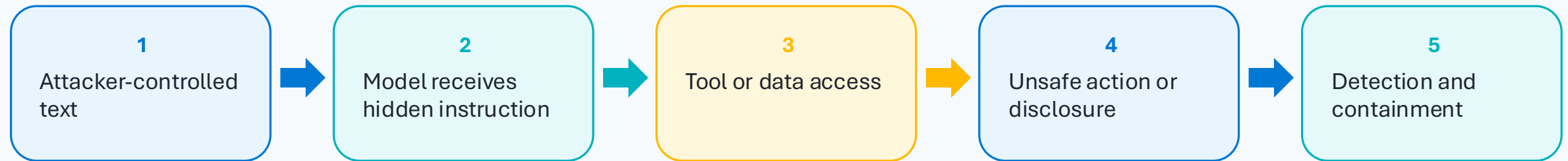
Key message

Useful AI should be valid, safe, secure, transparent enough and accountable.

AI systems expand the data and attack surface



Untrusted content can manipulate model behaviour



Key message

Treat model input and retrieved content as untrusted. Restrict tools, validate actions and require approval for high-impact operations.

Average accuracy can hide unequal outcomes

A Questions to test

- Who appears in the data?
- Who experiences errors?
- Which metric matters for each group?
- Can affected people challenge a result?



B Controls

- Representative evaluation data
- Group-level performance analysis
- Human review and appeal paths
- Continuous monitoring after release

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

Governance connects technical evidence to accountable decisions

A Operational controls

- Named owner and approved purpose
- Risk assessment and documentation
- Version, access and change management
- Incident response and retirement plan



B European context

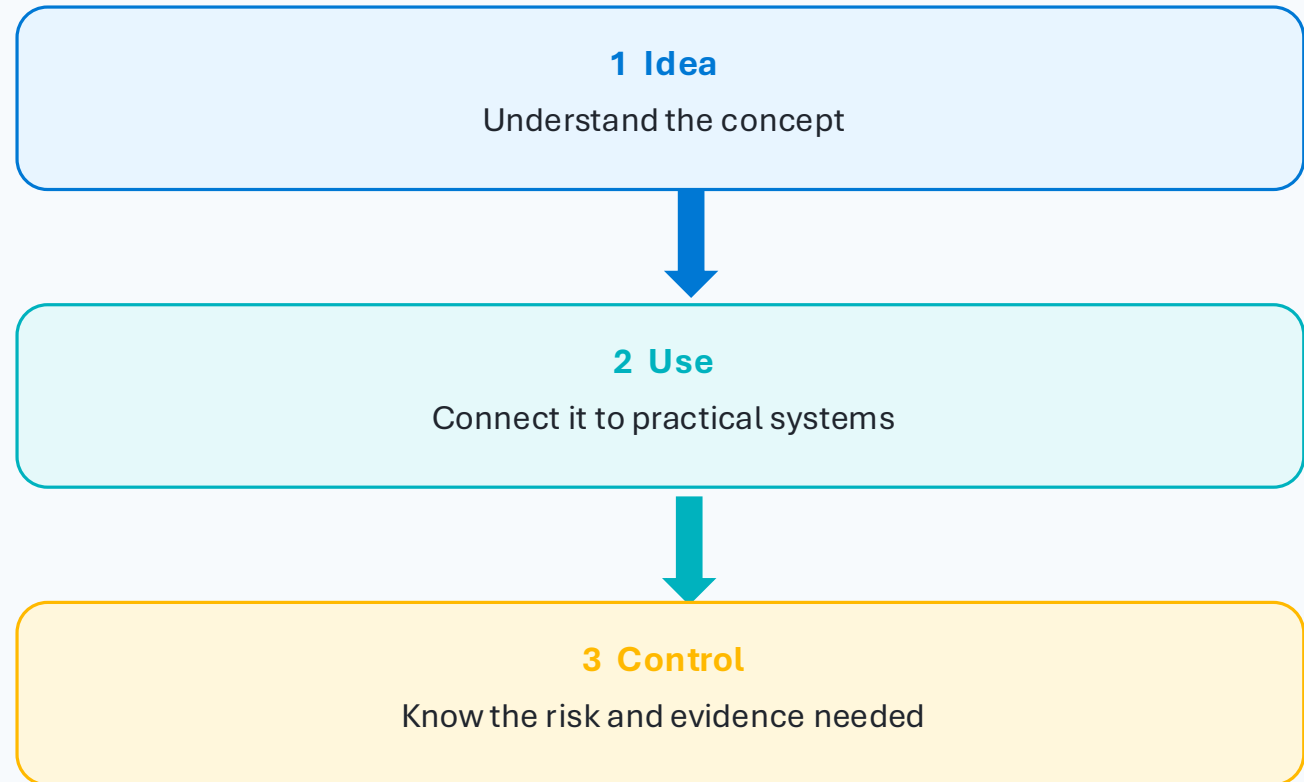
- The EU AI Act uses risk-based obligations
- Some practices face prohibitions
- High-risk systems require stronger controls
- Transparency duties apply to selected uses

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

Operations, limits and guardrails

Reliable AI depends on capacity, cost controls, evaluation and layered safety



TPM and RPM

A Tokens Per Minute (TPM)

- Limits the estimated token volume processed
- Includes prompt and requested output allowance
- Large contexts consume capacity quickly
- TPM also helps plan deployment throughput



B Requests Per Minute (RPM)

- Limits how many API calls arrive over time
- Short bursts can trigger throttling
- Clients should retry with backoff
- The TPM-to-RPM ratio varies by model

Decision lens

Choose based on the problem, the data type, explainability needs and the level of ambiguity.

Model cost

D Definition

Model cost is the total expense of running and operating the AI solution, not only the advertised token price.



M Mechanism

Cost depends on input and output tokens, model choice, retrieval, tool calls, hosting, storage, network use, evaluation and human review.



E Example

A summariser reduces cost by sending only relevant document sections to a smaller model and caching repeated answers.

Note

Track cost per successful task, not only cost per request. Include failure, retry and review costs.

Latency, throughput, caching and batching

D Definition

Latency measures how long one request takes. Throughput measures how much work the system completes over time.



M Mechanism

Caching reuses safe repeated results. Batching groups work for efficient processing. Streaming shows partial output before completion.



E Example

A nightly document-indexing job uses batches, while a live support assistant streams its answer for a faster user experience.

Note

Optimising speed can affect freshness, cost, privacy and output quality.

AI guardrails

D Definition

Guardrails are layered controls that constrain inputs, outputs, tool use and workflow decisions so the system stays within approved boundaries.



M Mechanism

Controls can include content filters, allow-lists, structured schemas, business rules, identity checks, approvals and runtime monitoring.



E Example

A finance agent may draft a transfer request but cannot submit it until a manager approves the amount and destination.

Note

Guardrails reduce risk but cannot guarantee safety. Teams still need evaluation, incident response and accountability.

Azure AI Content Safety

D Definition

Azure AI Content Safety analyses text and images for harmful or unwanted material and provides features for prompt attack and groundedness risks.



M Mechanism

Applications call moderation APIs before or after model generation and apply thresholds or review routes based on the returned categories and severity.



E Example

A community application blocks severe hateful content and sends uncertain cases to moderators instead of publishing automatically.

Note

Content Safety is one guardrail layer. Domain rules, access control and human review remain necessary.

Structured output and validation

Definition

Structured output constrains a model response to a defined shape, such as JSON fields with allowed types and values.



Mechanism

The application requests a schema, validates the result and rejects or repairs output that does not meet business rules.



Example

An invoice assistant must return supplier, currency and total as typed fields before downstream code can use them.

Note

A valid structure can still contain wrong facts, so content validation remains necessary.

Evaluation datasets and regression tests

Definition

An evaluation dataset is a versioned set of representative inputs, expected properties and scoring rules used to compare system versions.



Mechanism

Teams run the same cases after changing a model, prompt, retrieval index or tool and investigate any quality or safety regression.



Example

A support agent test set includes normal questions, ambiguous requests, restricted-data attempts and known prompt injections.

Note

Production feedback should add new failure cases without exposing sensitive user data.

Tracing and observability

D Definition

Observability records enough evidence to understand what the AI system did, why it took time or cost money and where a failure occurred.



M Mechanism

A trace can link the user request, prompt version, retrieved sources, model response, tool calls, latency, token use and final outcome.



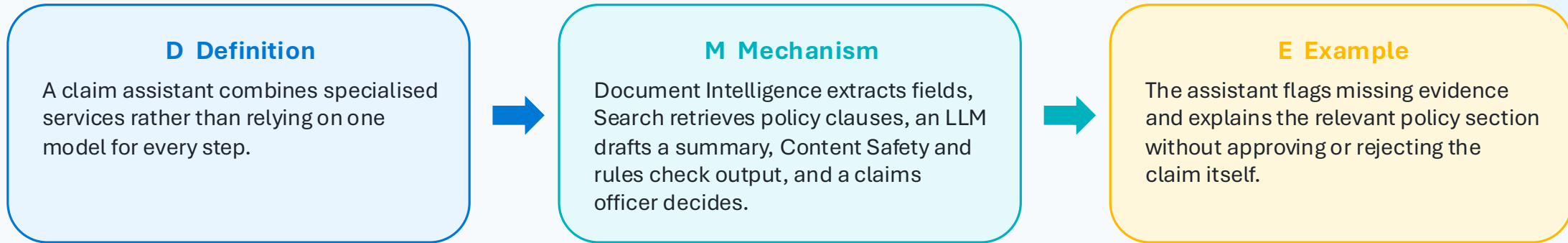
E Example

An engineer discovers that slow answers come from repeated search calls rather than from model inference.

Note

Logs require privacy controls, retention rules and access restrictions because they may contain sensitive content.

Generative AI example: insurance claim assistant

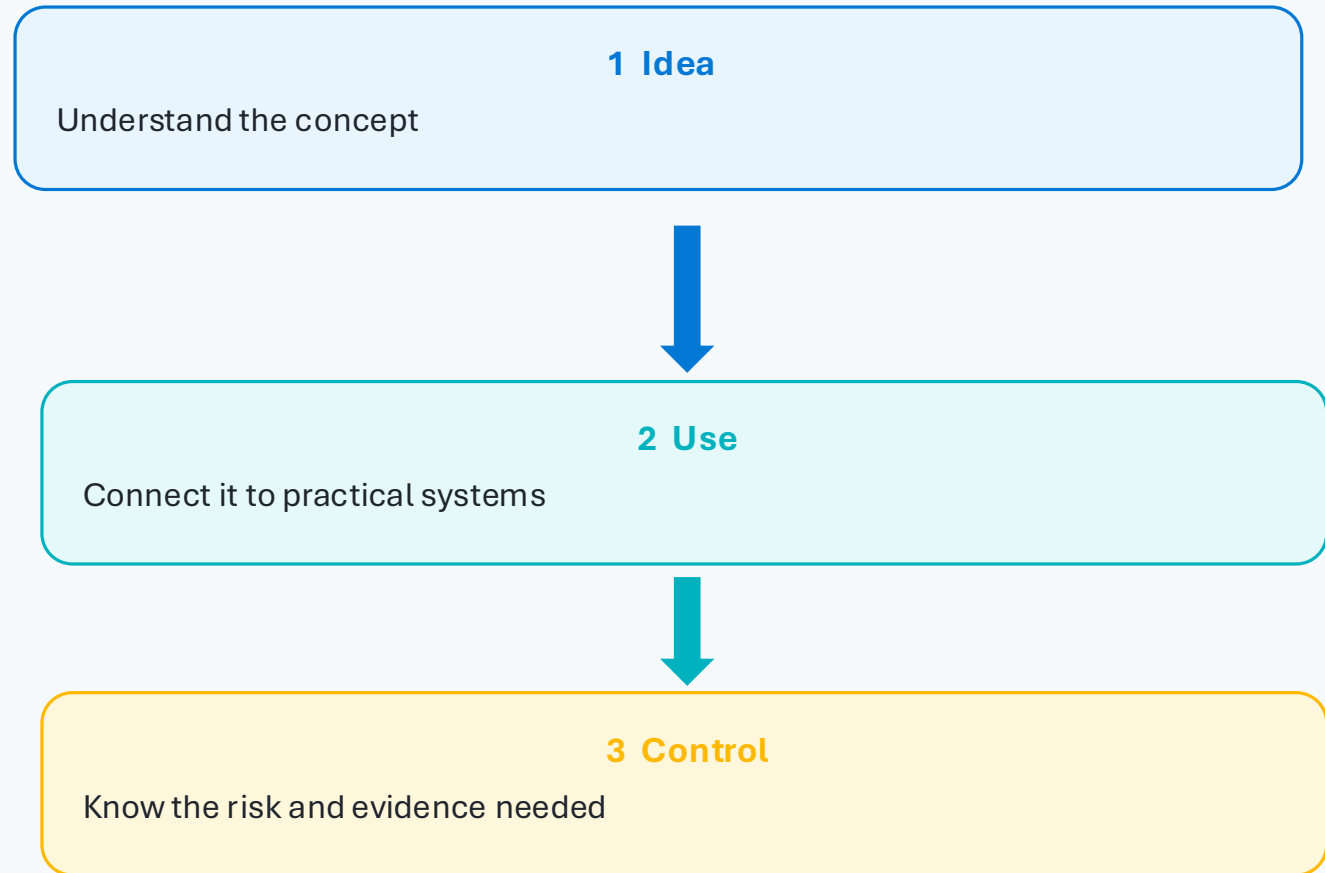


Note

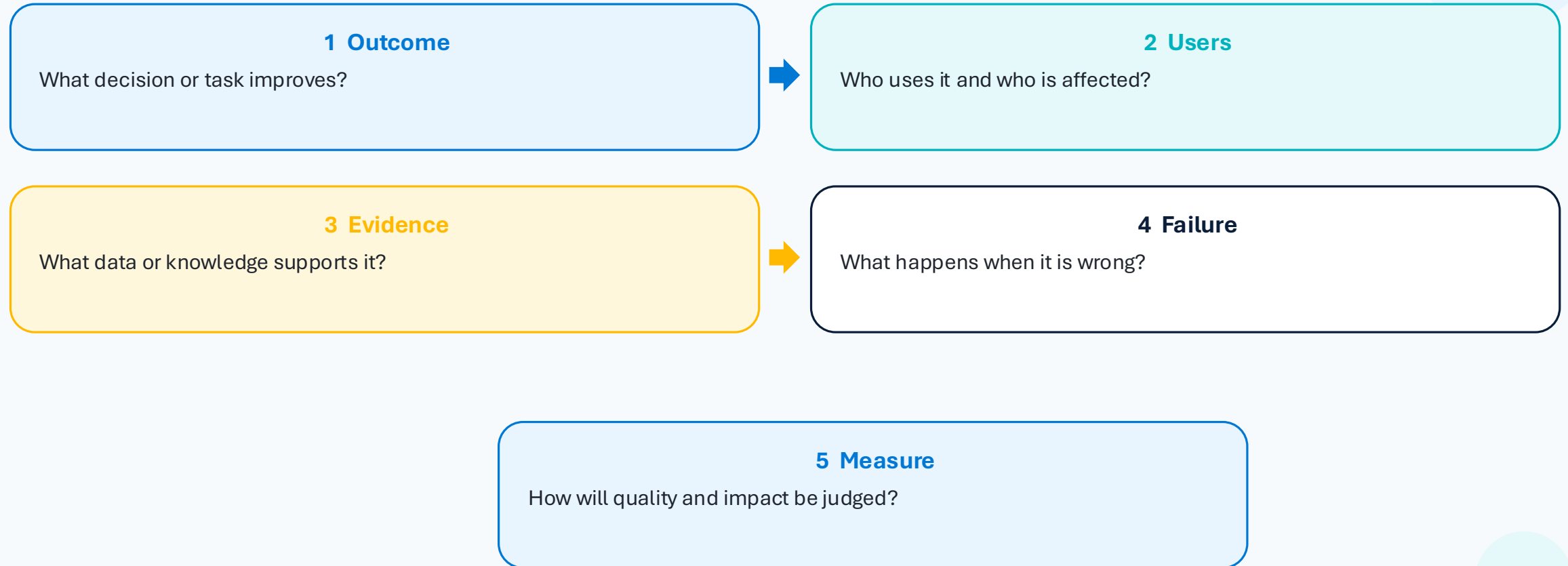
This design separates extraction, retrieval, generation, validation and human accountability.

Applying the ideas

Start with the user problem, then choose the smallest dependable system



A useful AI brief answers five questions



Tool choice follows constraints

1

Prefer rules or conventional software when the process is stable and explicit

3

Use generative models for flexible language, media or code tasks

5

Add agents only when the workflow needs flexible multi-step tool use

2

Use classical ML for repeatable prediction from structured data

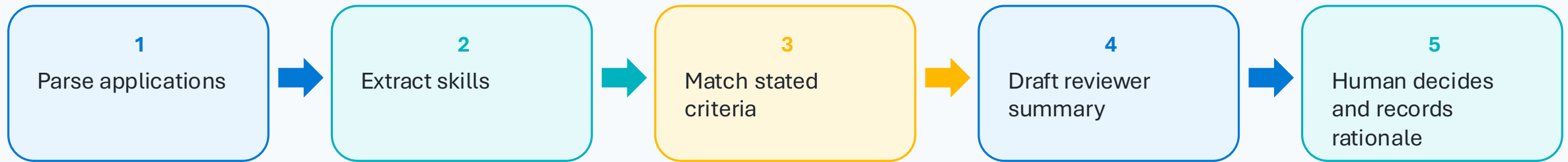
4

Add retrieval when answers must use current or private knowledge

6

Increase human oversight as consequence, uncertainty or sensitivity rises

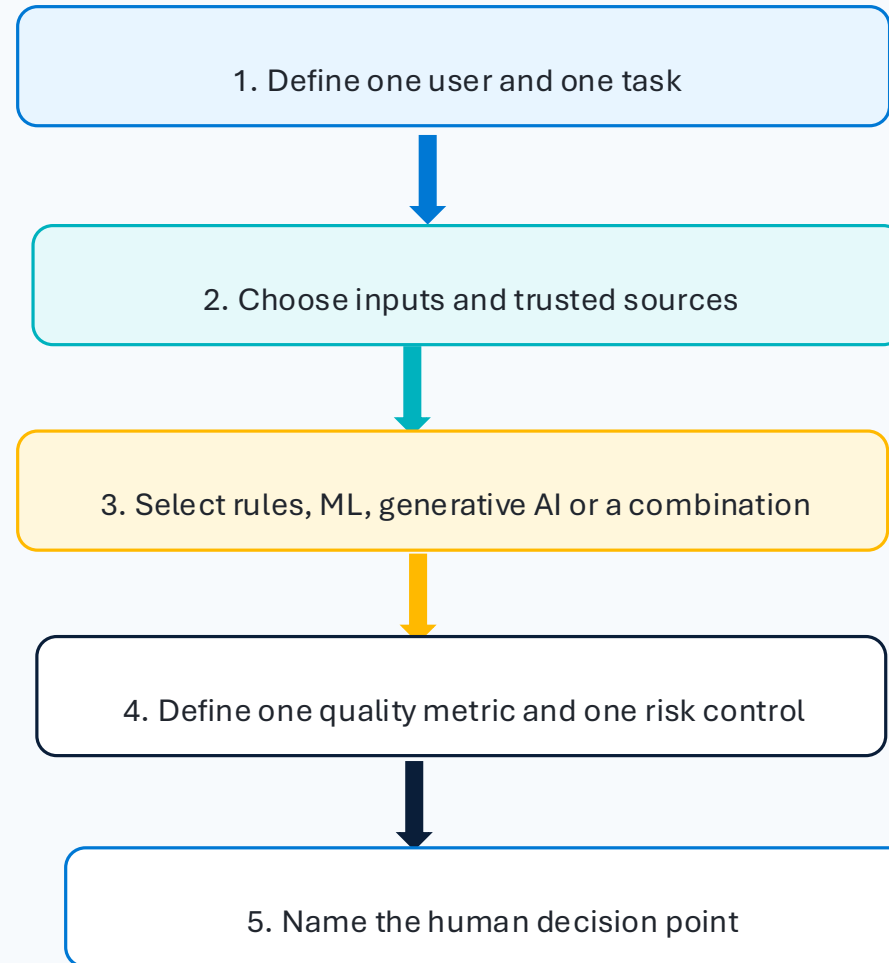
Recruitment assistant: one problem, several AI choices



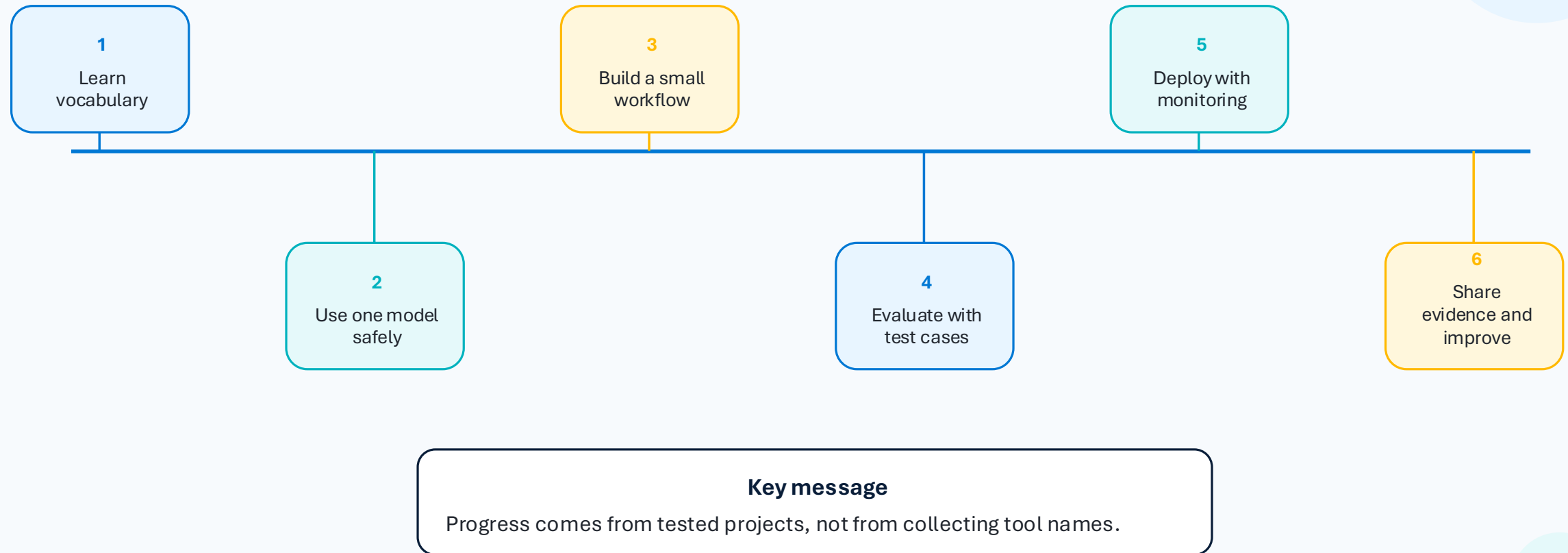
Key message

Do not infer protected traits, hide ranking logic or let the model make the final employment decision.

Design an AI solution on one page



Build AI literacy through repeated practice



AI is a system of choices

1

Start with purpose and affected people

3

Match the method and tool to the task

5

Design security, oversight and accountability before deployment

2

Treat data, evaluation and operations as core design work

4

Ground generative systems in evidence and validation

6

Keep learning through labs, documentation and measured experiments

Q & A / WRAP-UP

Questions and discussion

Open discussion

Invite questions, comments and reflections from the session.

**Starter question**

What part of the AI landscape do you want to explore first?

Closing action

Use the lab repository to practise the concepts after the session.